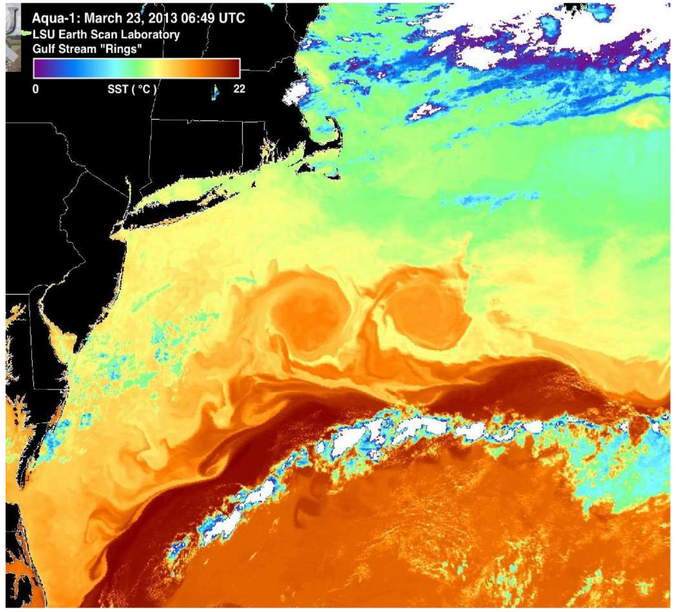
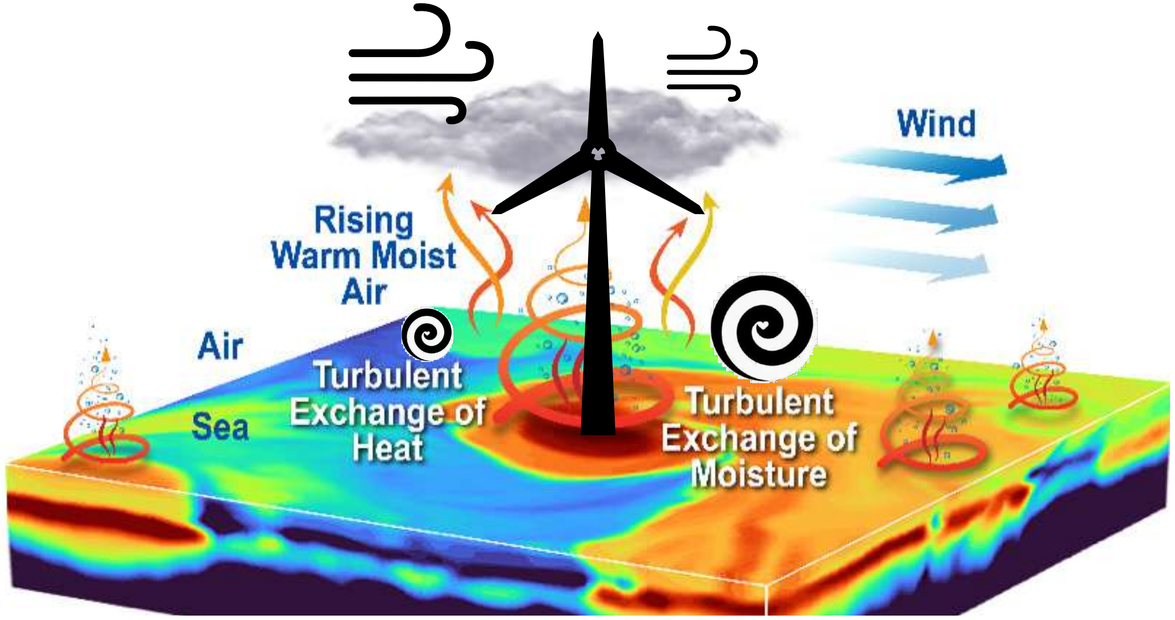
Stratified Shelf Ocean and Fetch-Dependent Sea State Responses to Large-Scale Offshore Wind Farms on the US East Coast







SNU Department Seminar
June 25, 2026

Hyodae Seo & César Sauvage, Christoph Renkl (UH),
Julie Lundquist (JHU/NREL),
Anthony Kirincich, Jim Edson, Carol Anne Clayson (WHOI)
& Seth Zippel (OSU)

University of Hawai'i Mānoa

U.S. Department of Energy | Energy Efficiency & Renewable Energy

Woods Hole Oceanographic Institution

RESEARCH ARTICLE | ATMOSPHERIC SCIENCE

Sea surface warming and ocean-to-atmosphere feedback driven by large-scale offshore wind farms under seasonally stratified conditions

HYODAE SEO, CÉSAR SAUVAGE, CHRISTOPH RENKL, JULIE K. LUNDQUIST, AND ANTI

JGR Oceans

Research Article

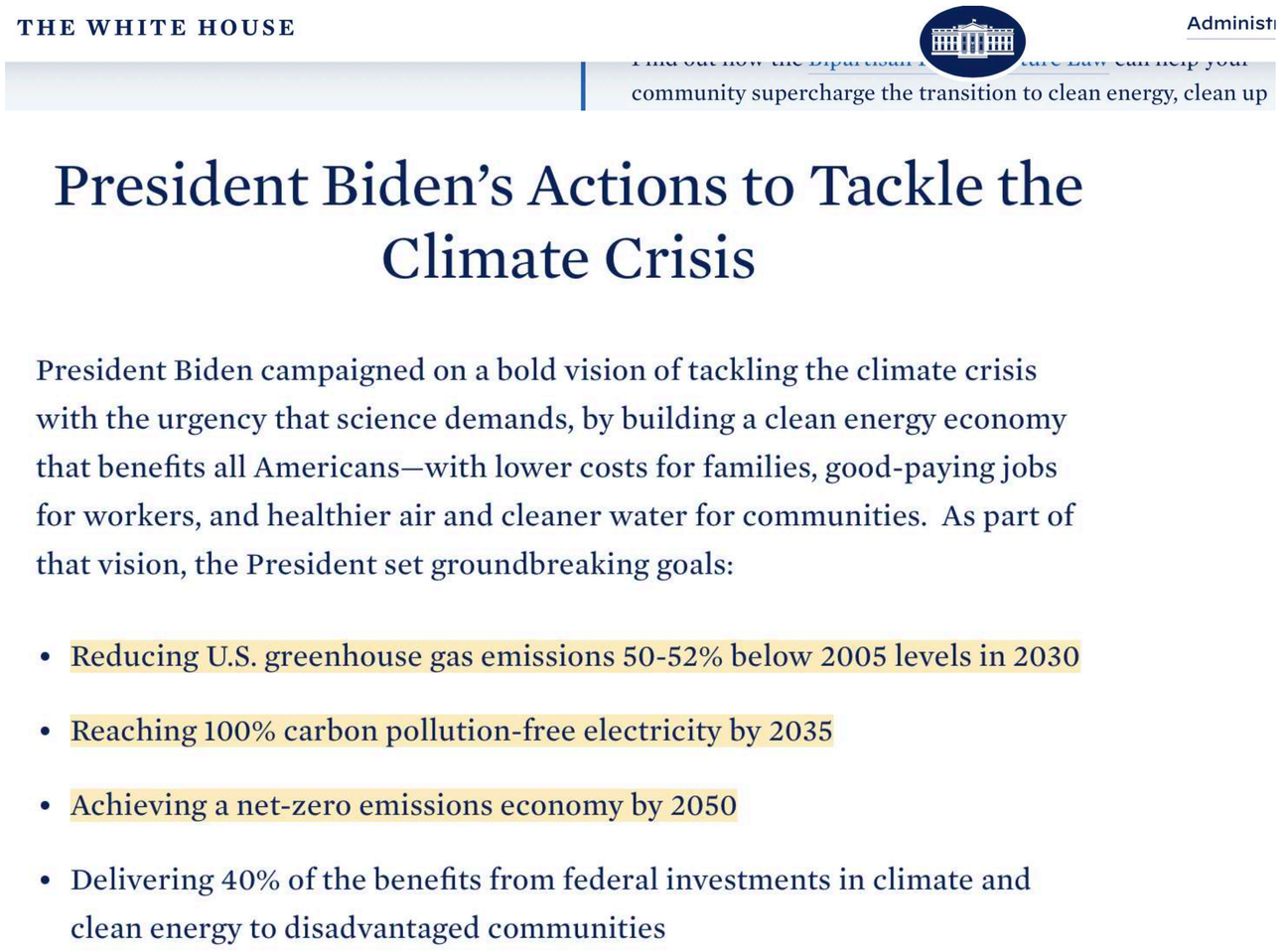
Fetch-Dependent Surface Wave Responses to Offshore Wind Farms in the Northeast U.S. Coast

César Sauvage, Hyodae Seo, Seth Zippel, Carol Anne Clayson, James B. Edson

First published: 23 December 2025 | https://doi.org/10.1029/2025JC023156 | VIEW METRICS

Accessibility issue? Request accessibility update.

Before and After:
US Federal Government's Climate Action Plans





- 30 GW of offshore wind energy by 2030 (2.5K wind turbines)
- 3000 GW by 2050 (250K)

https://www.whitehouse.gov/climate/#:~:text=Reducing U.S. greenhouse gas emissions,zero emissions economy by 2050

https://www.whitehouse.gov/presidential-actions/2025/01/temporary-withdrawal-of-all-areas-on-the-outer-continental-shelf-from-offshore-wind-leasing-and-review-of-the-federal-governments-leasing-and-permitting-practices-for-wind-projects/

Big Picture Questions

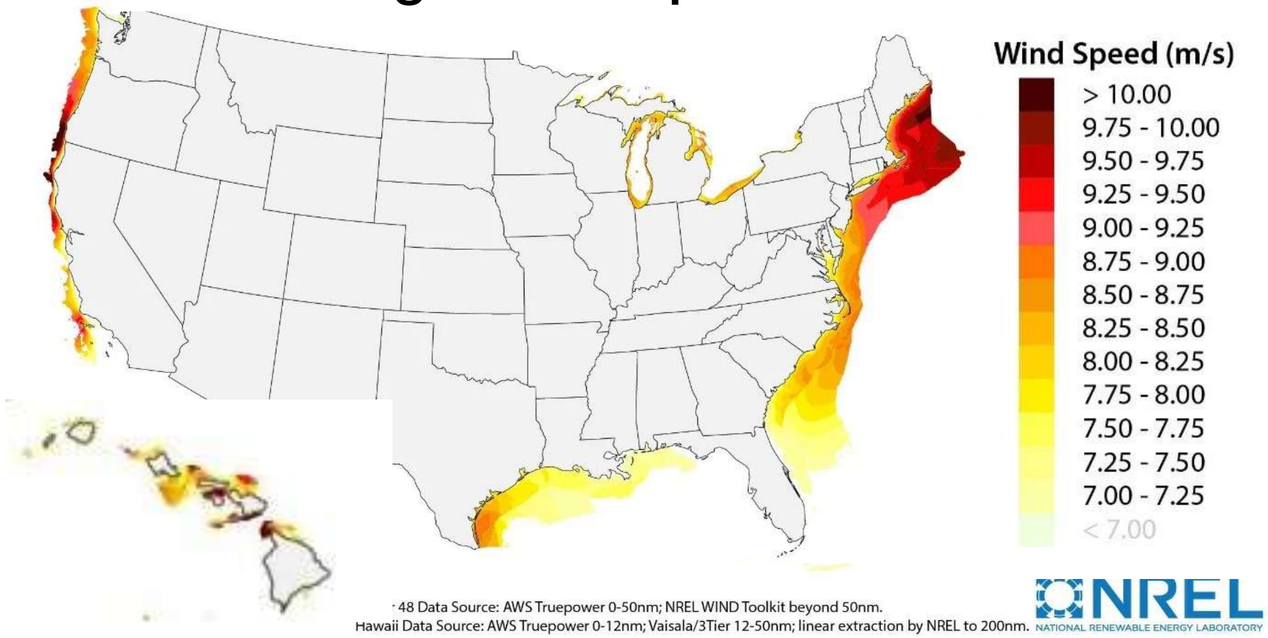
1. Why offshore wind?

2. Why does air-sea interaction matters for wind energy production?

3. What are the environmental impacts of wind energy production?

Why offshore wind?

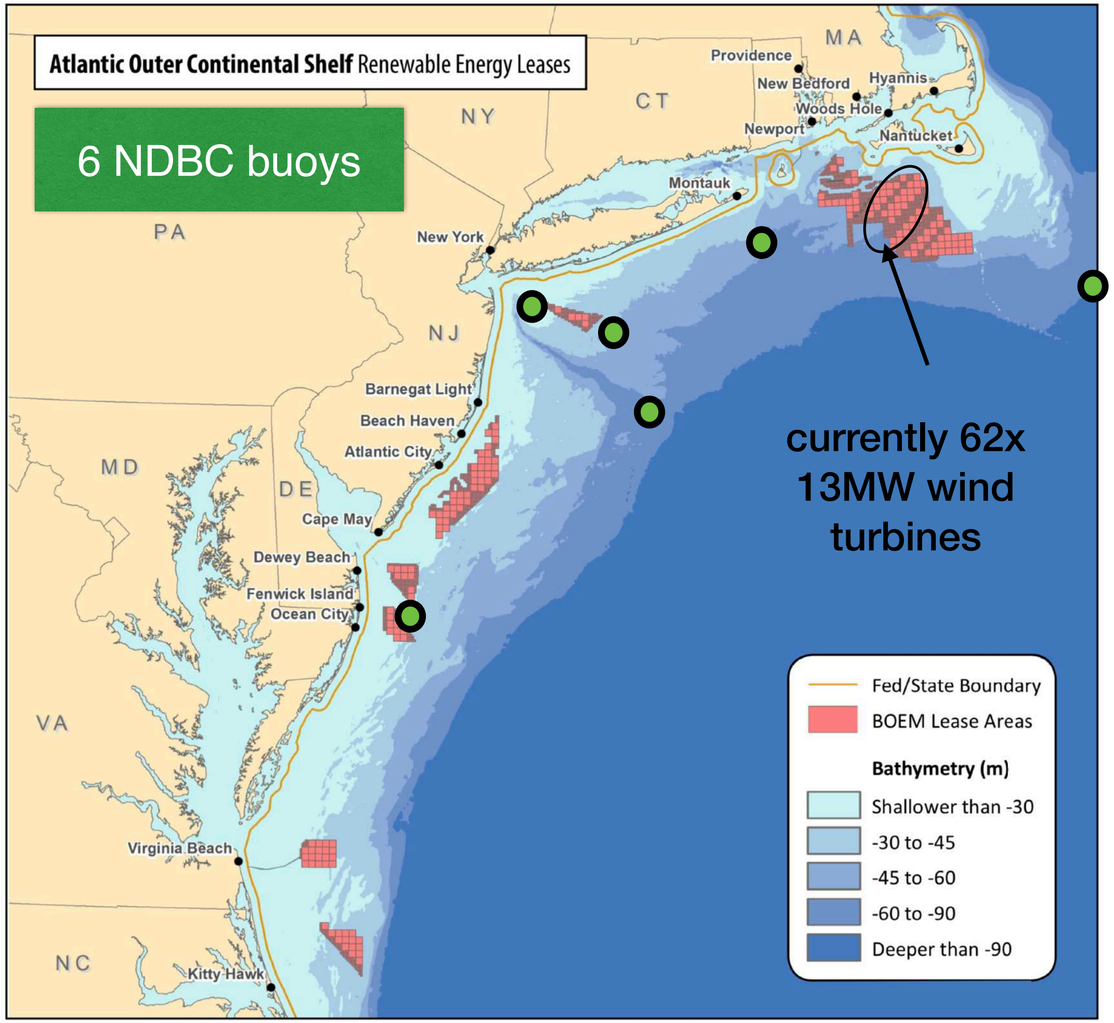
hub height wind speeds



- Winds are stronger and steadier offshore
- Supply is higher offshore, closer to cities with high electricity demand
- Less conflict with local communities

- Previous Federal Gov. Climate Action Plan: 30 GW by 2030 & 3000 GW by 2050
- Protecting biodiversity & promoting ocean co-use
- Huge changes in the way we use coastal/shelf oceans

US Northeast Lease/call areas (NREL 2022)



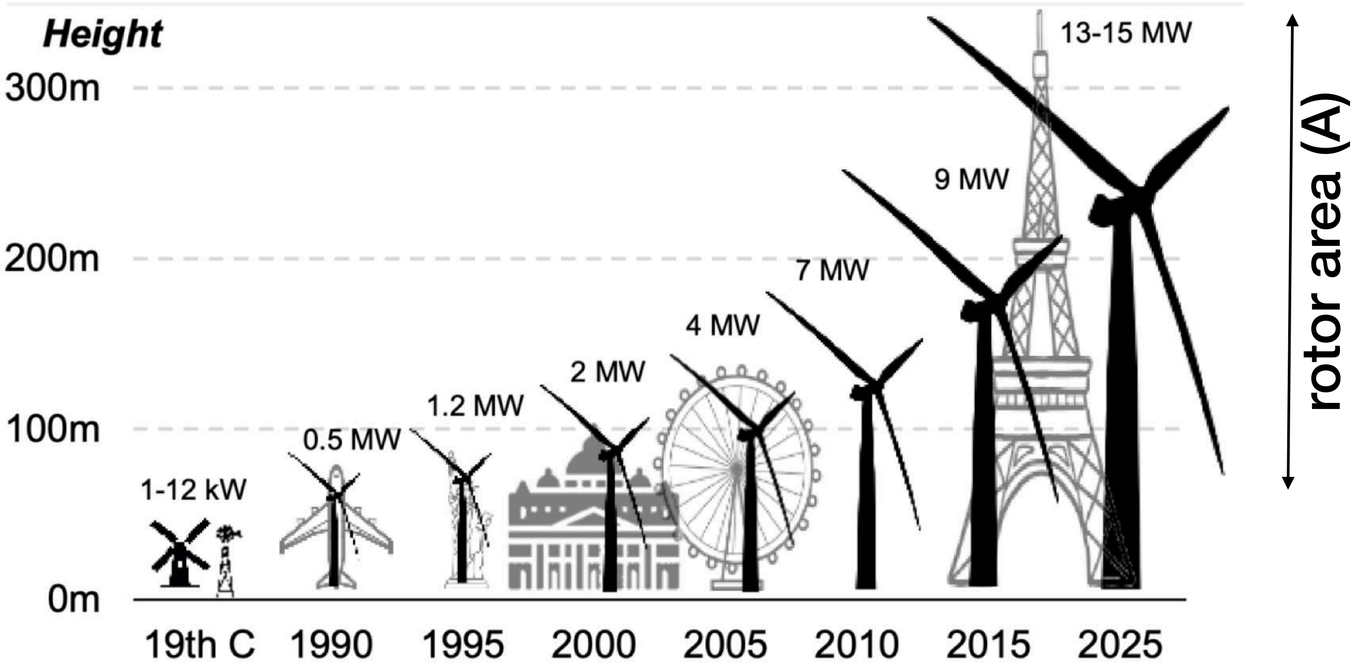
Big Picture Questions

1. Why offshore wind energy?

2. Why does air-sea interaction matters?

3. What are the environmental impacts of wind energy production?

Wind turbines are huge and getting bigger



Source: IEA (2019) Offshore Wind Outlook 2019

rotor area

$$P = \frac{1}{2}\rho A V^3$$

wind speed

- Big turbines drive cost down / MW.
- Engineering challenges in transportation, installation, construction, and maintenance
- Greater environmental impacts



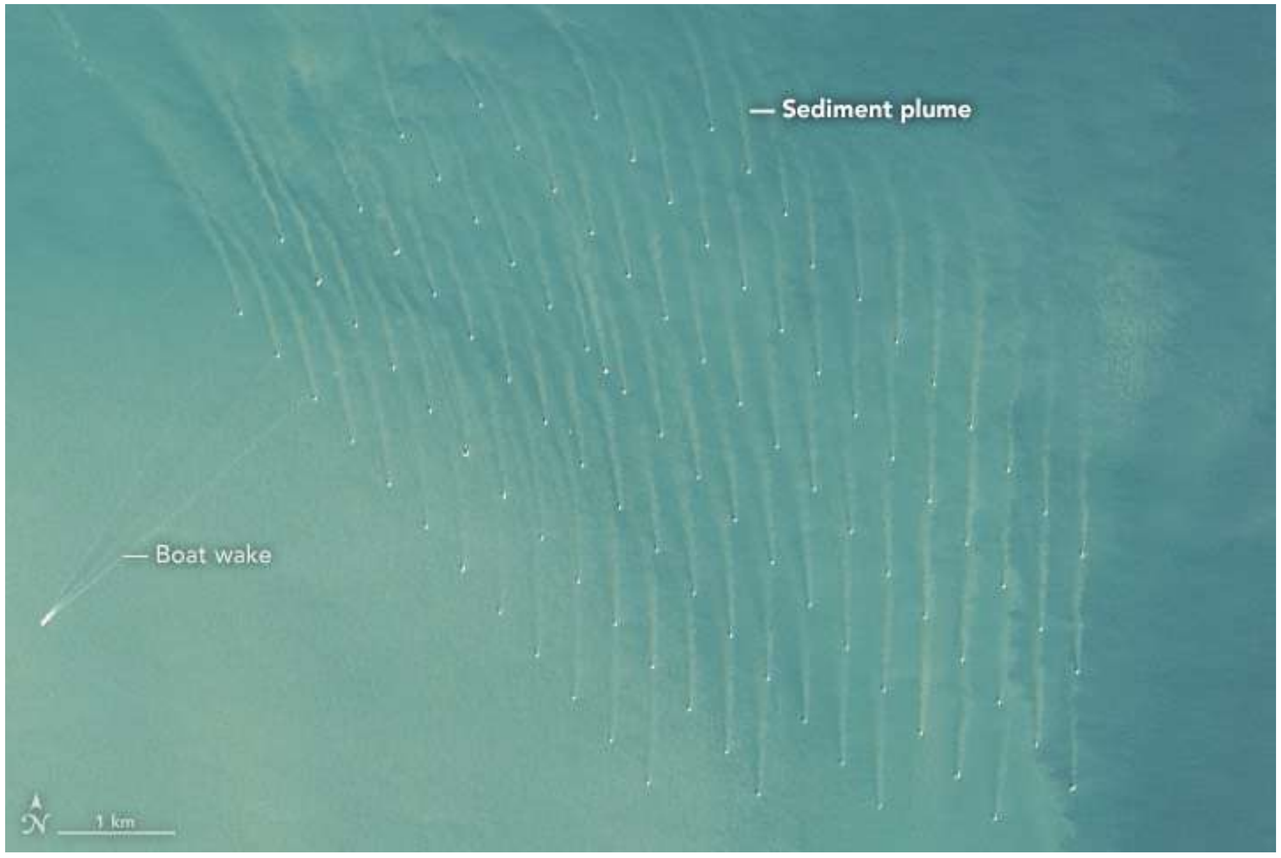
Wind Wakes

Wind turbines extract kinetic energy from the atmosphere to generate electricity. This reduces wind speed and increases turbulence downstream.

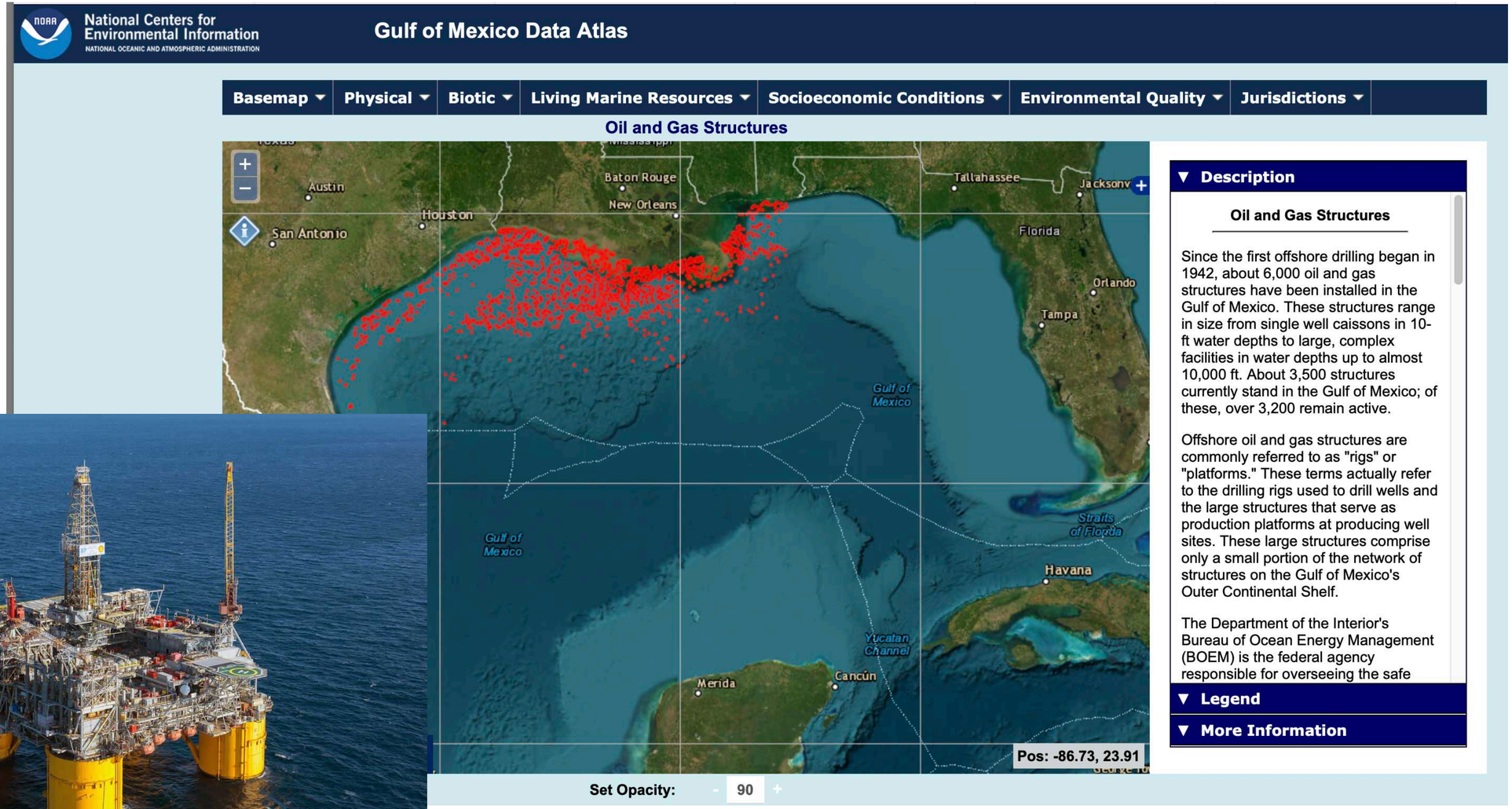
Wind Wakes

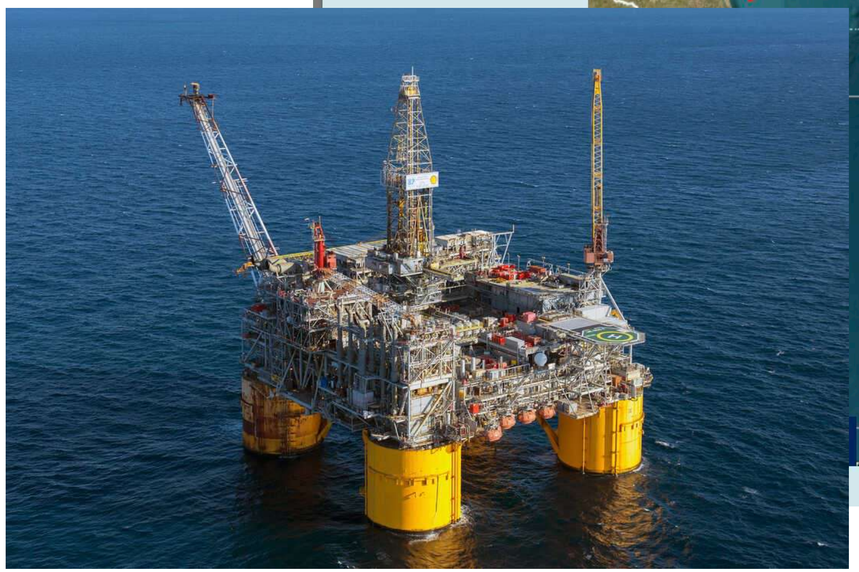


Ocean Wakes



High-density offshore structures





Currently, there are ~3,200 active oil rigs in the Gulf of Mexico.
However, unlike wind farms, they do not generate significant structural wind wake impacts.

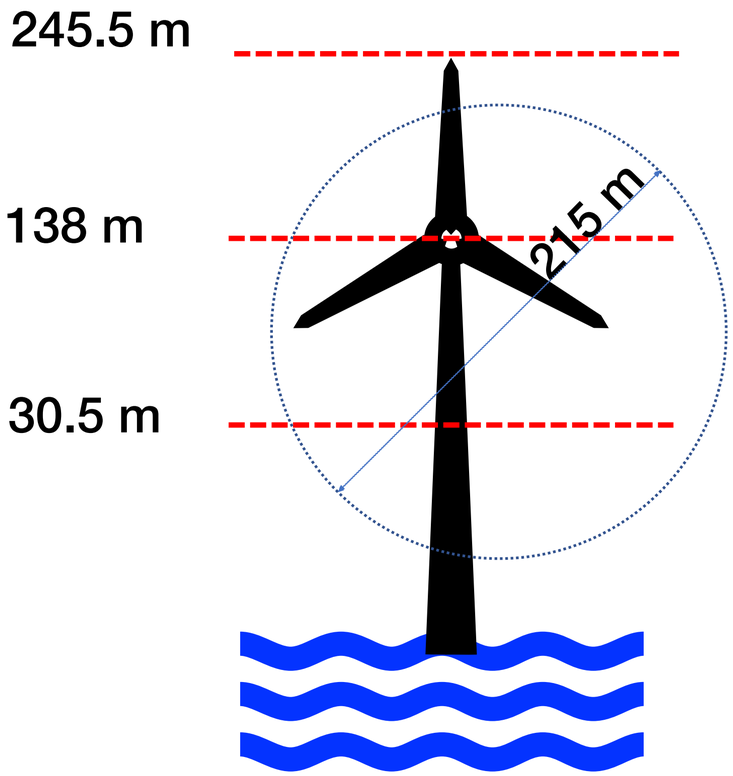
Big Picture Questions

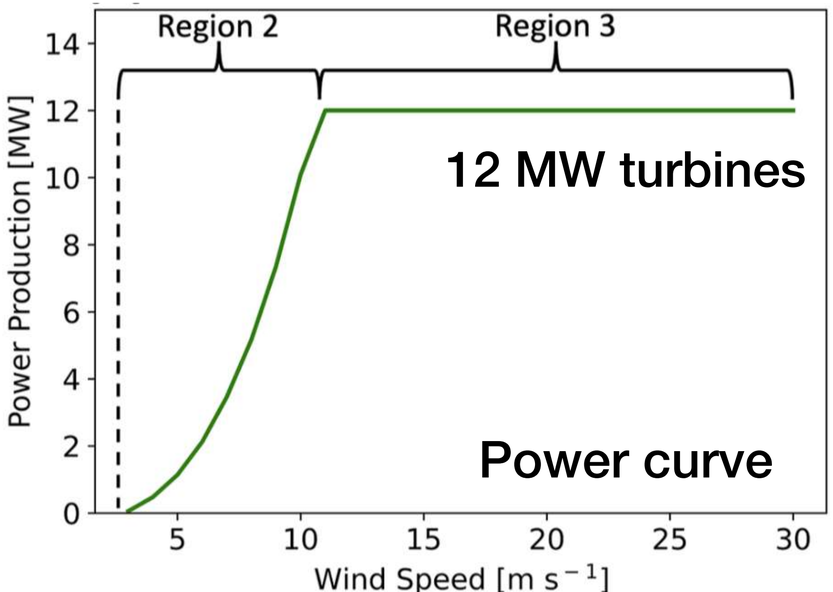
1. Why offshore wind energy?

2. Why does air-sea interaction matters for wind energy production?

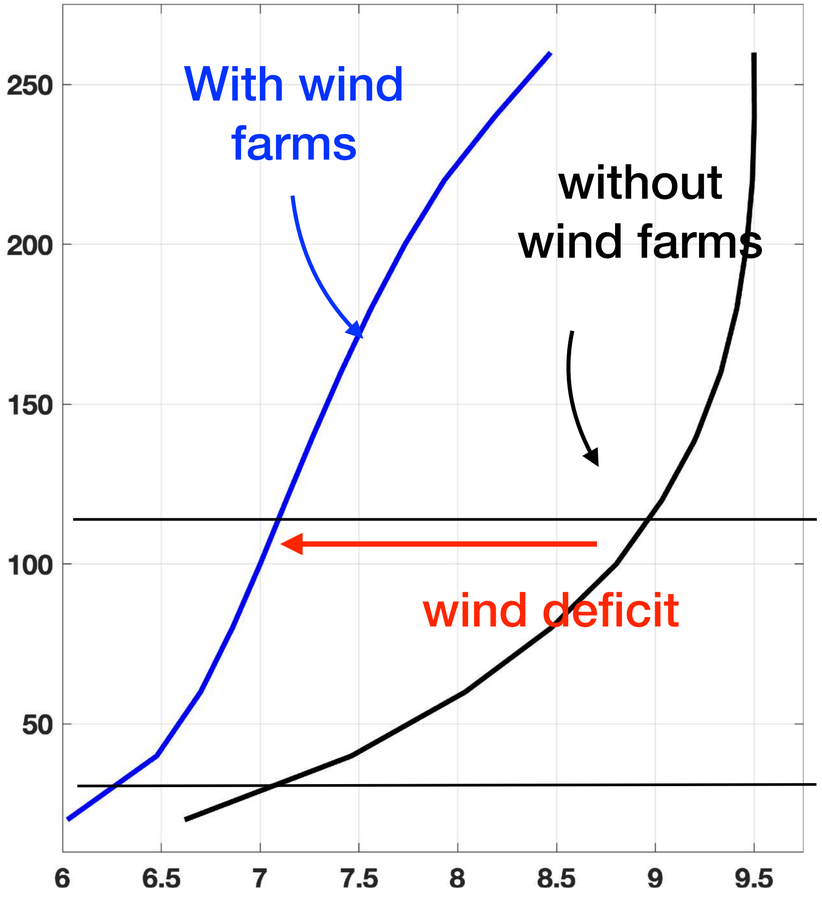
3. What are the ocean impacts of wind energy production?

Wind farm wakes in mesoscale models

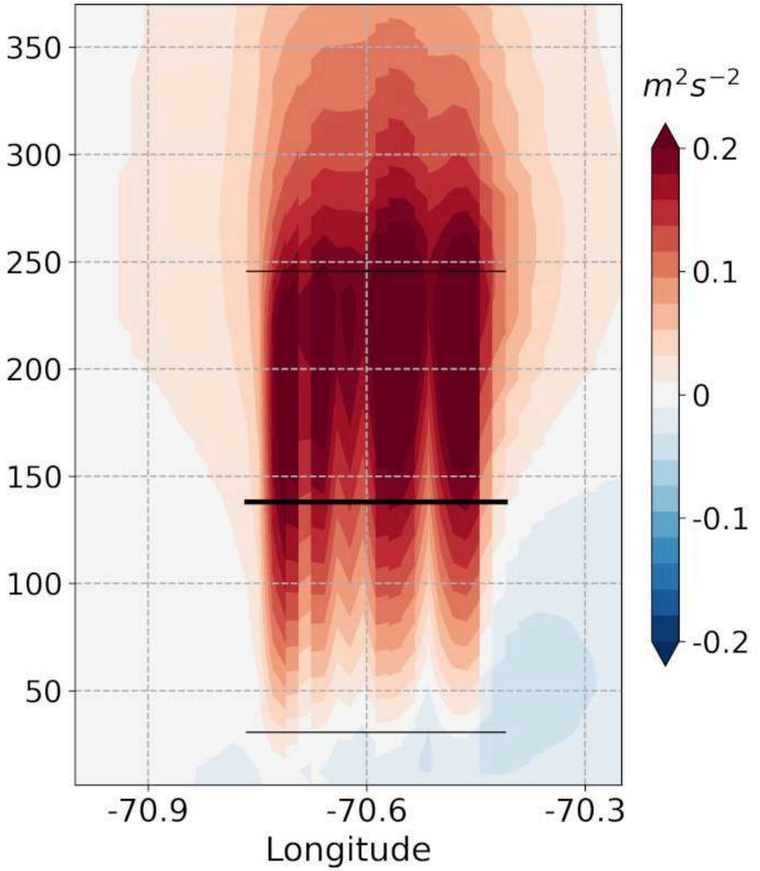




Wind Speed Changes



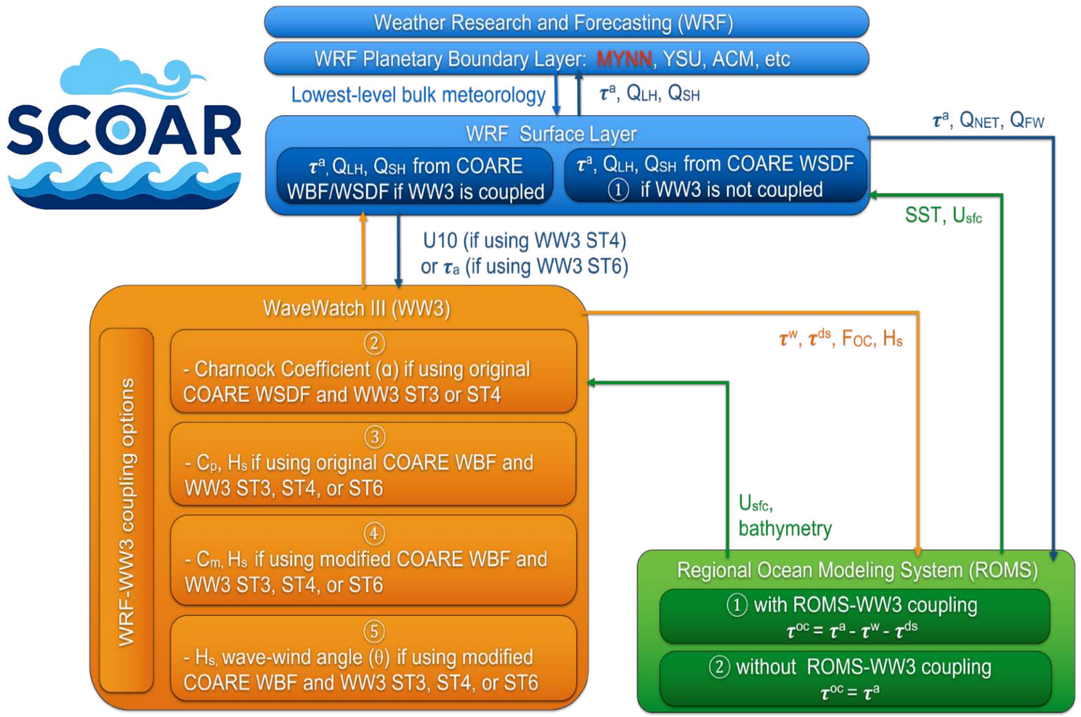
TKE Changes



Fitch et al. (2012)

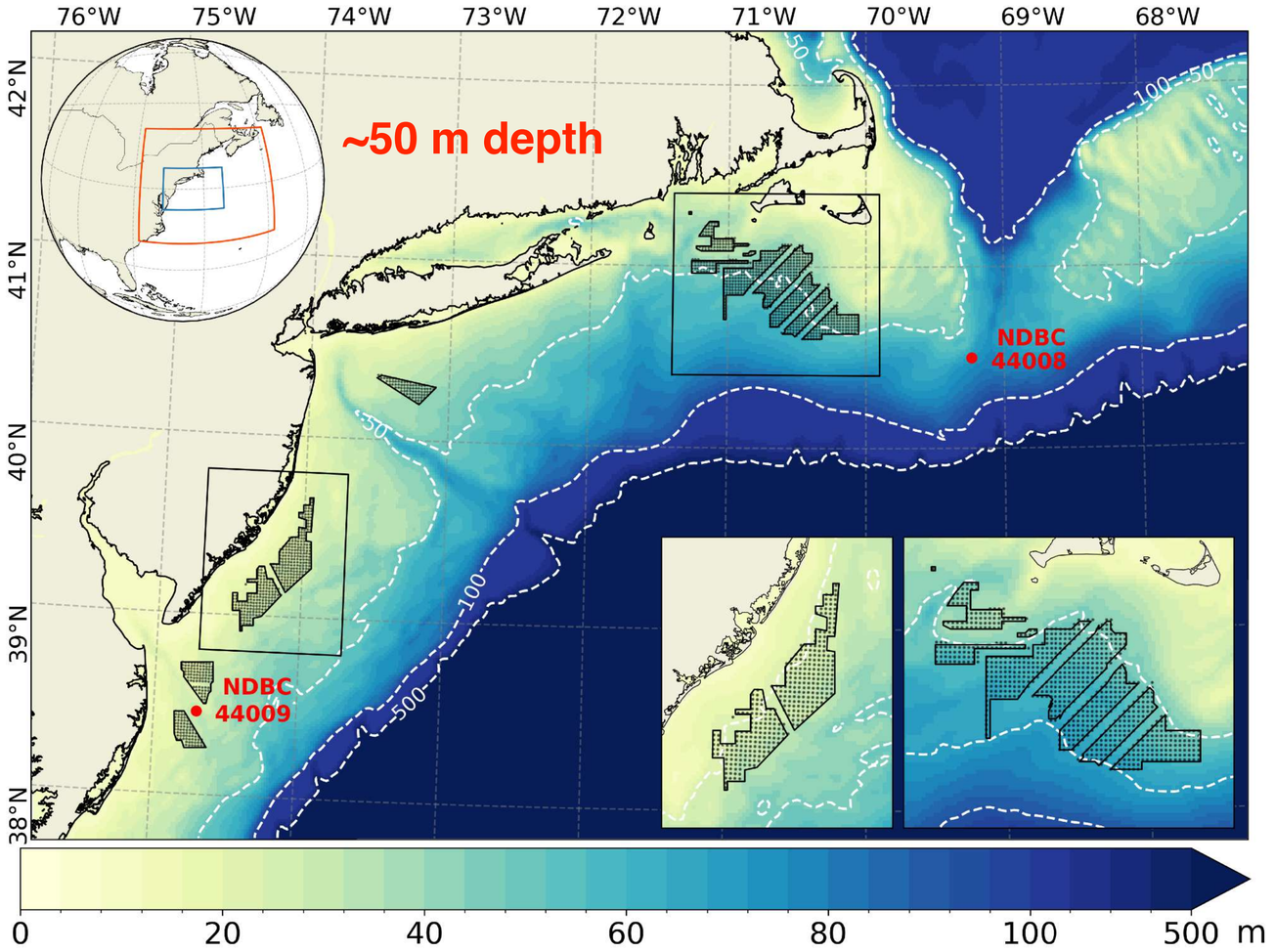
Wake-ocean interactions in a fully coupled mesoscale model

SCOAR Regional Coupled Model
Seo et al. (2007, 2016), Sauvage et al. (2024)



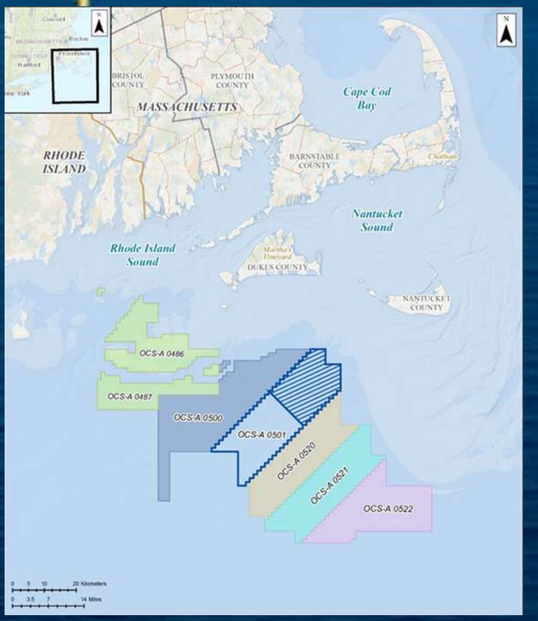
Experiment	Wind Farm (WF)	Ocean Coupling (OC)
OC_WF	Yes	Yes
OC_NWF	No	Yes
NOC_WF	Yes	No

1.5 km SCOAR WRF-ROMS-WW3 simulations



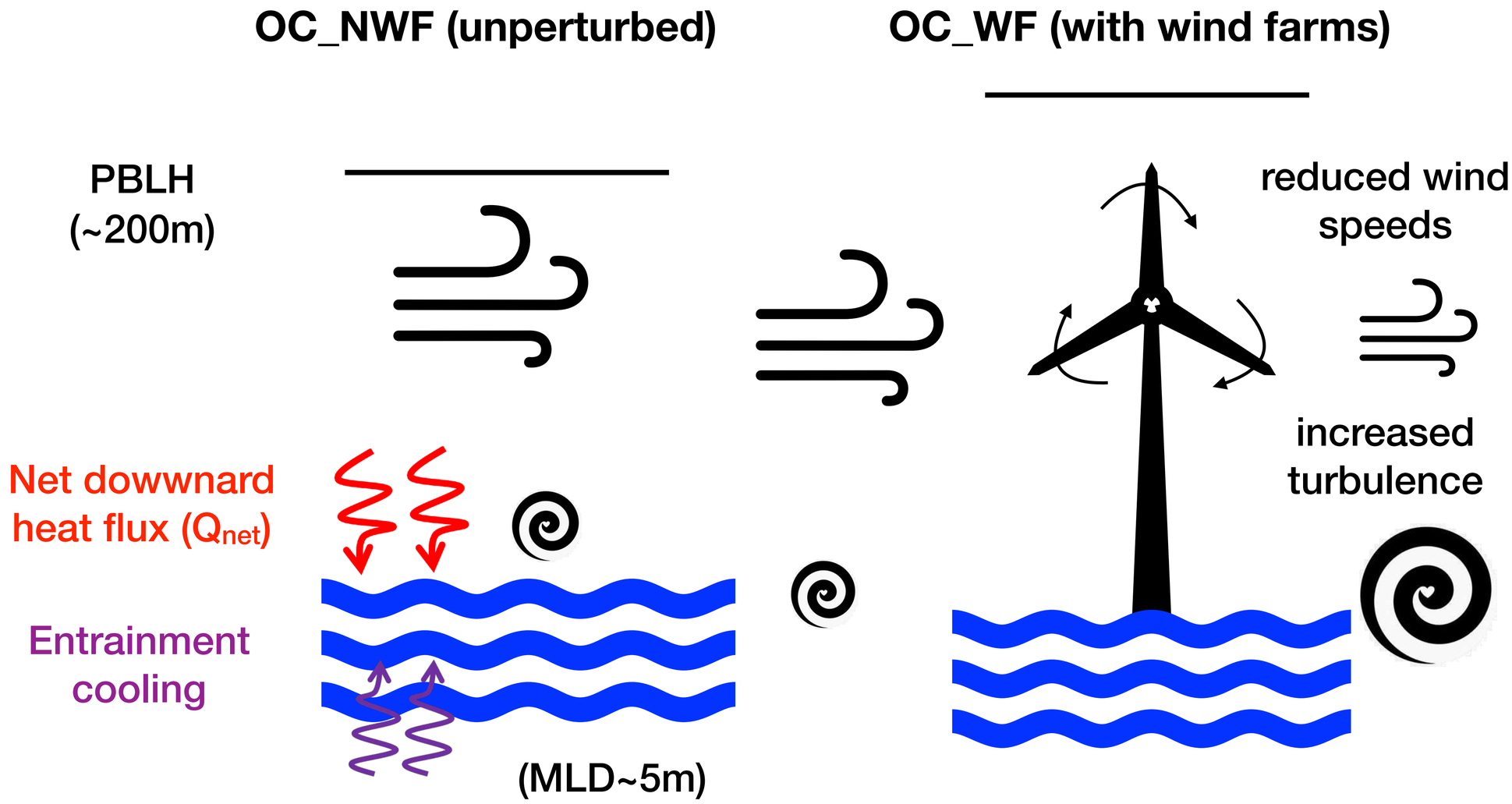
1418 X 12MW wind turbines in all MAB lease areas, with total installed capacity = 17016 MW

1 nmi.



Vineyard Wind 1: 62x 13 MW turbines with a total capacity of 806 MW. → Largest completed offshore wind project in the US

Wake-ocean interactions

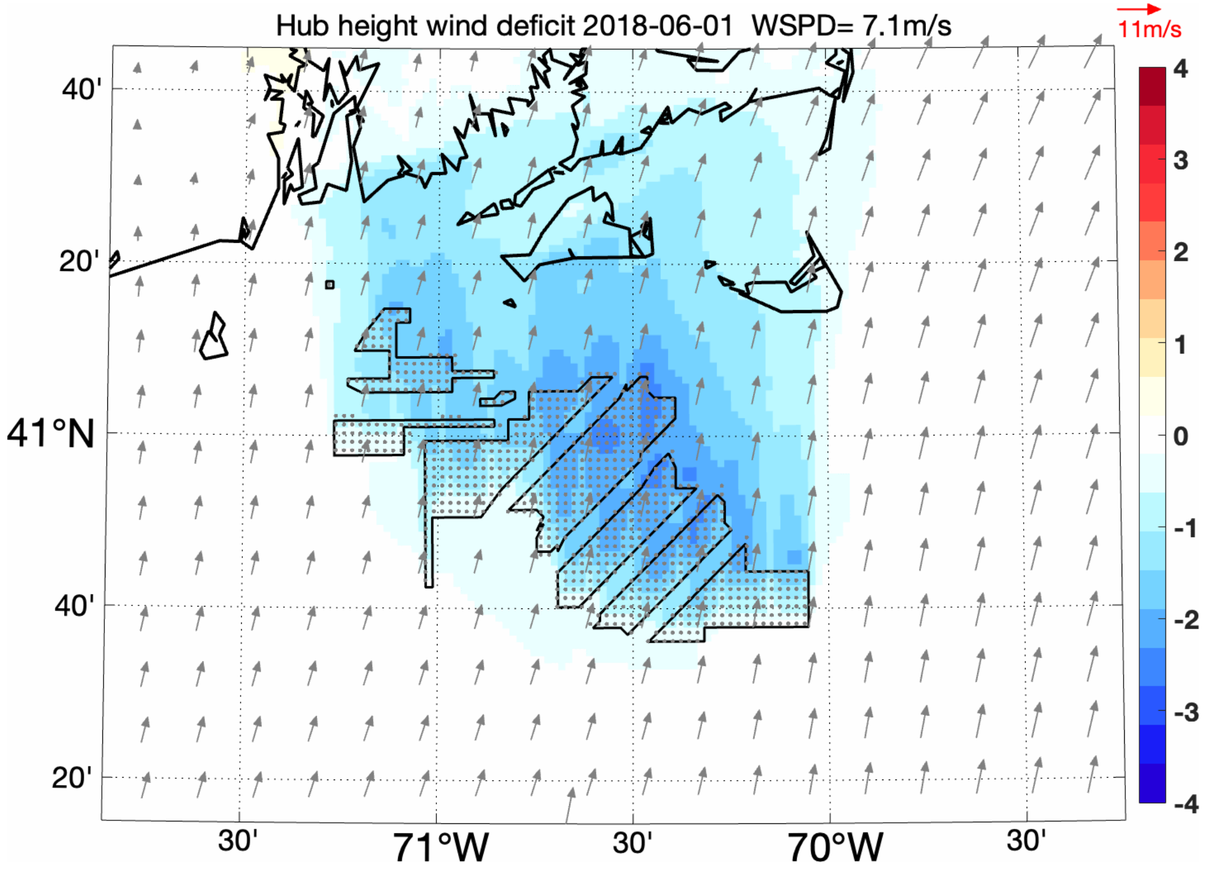


Diagnose the impacts on air-sea interactions induced by wind wake effects

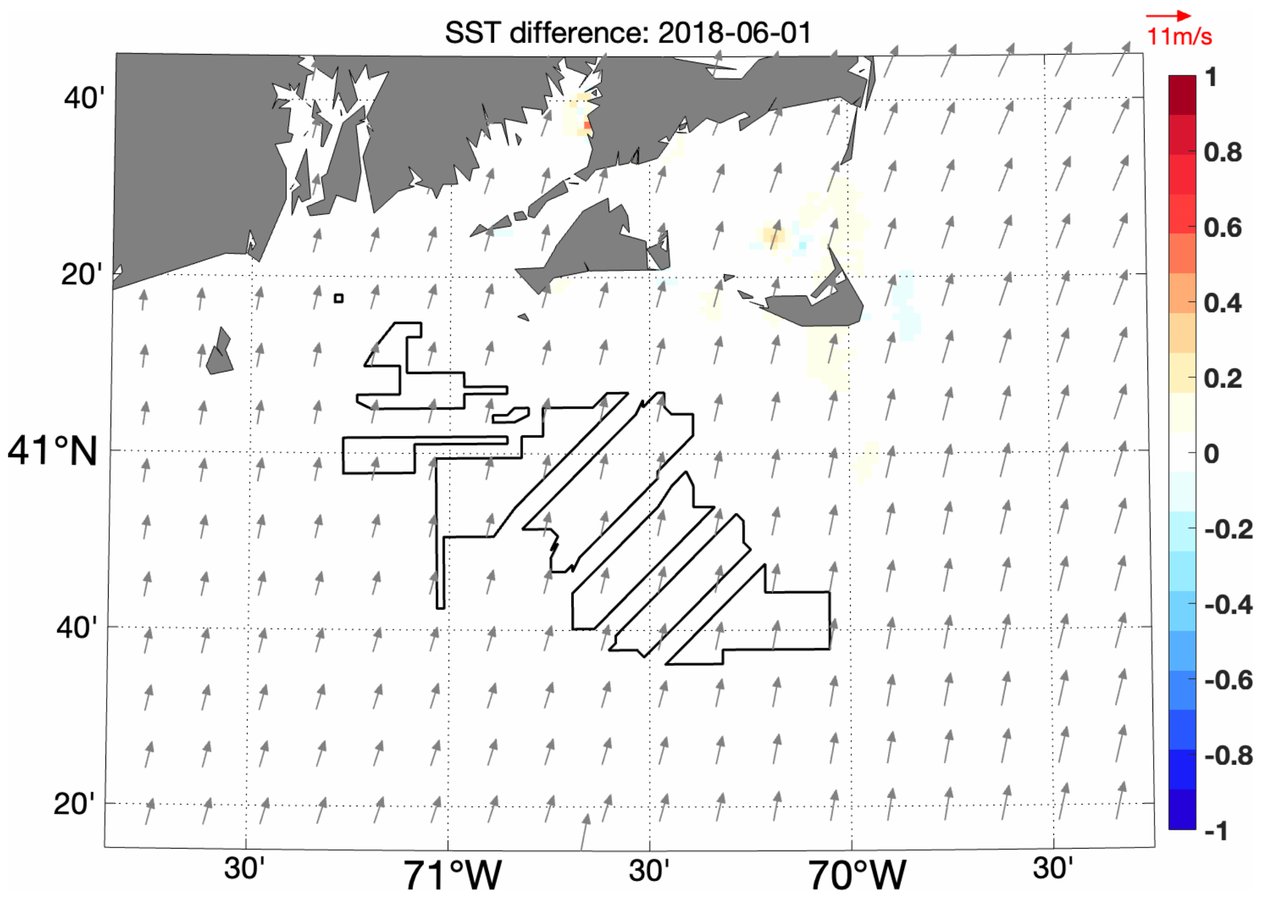
Parameterized wake effects in the coupled model:

OC_WF - OC_NWF

Wind deficit @ hub height (138 m)



Sea surface temperature response

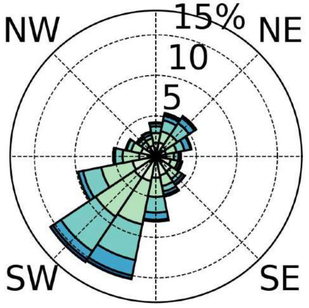


>11m/s: rated speed

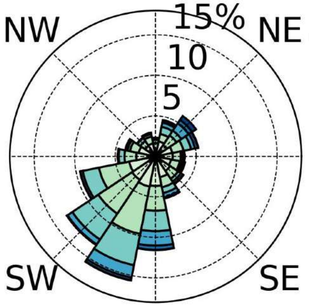
JJA Climatology

Wind

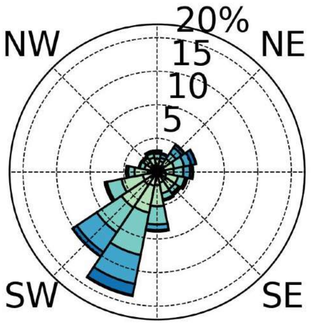
NOWF



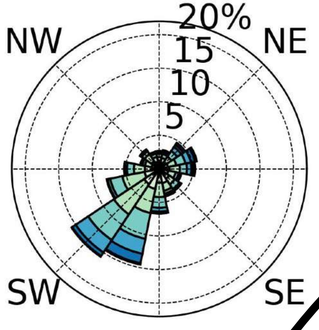
NDBC 44008



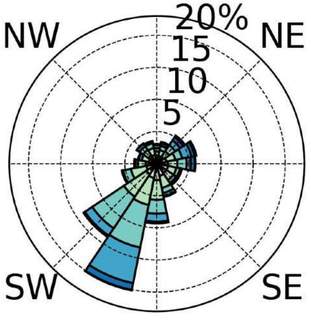
NOWF



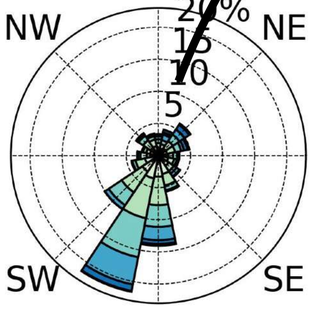
NDBC 44025



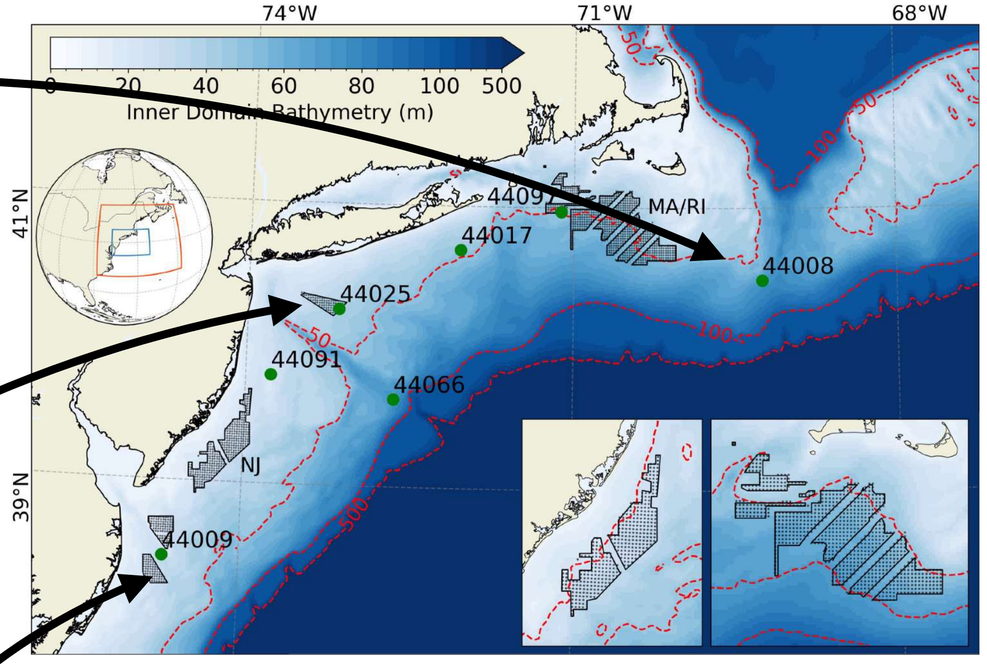
NOWF



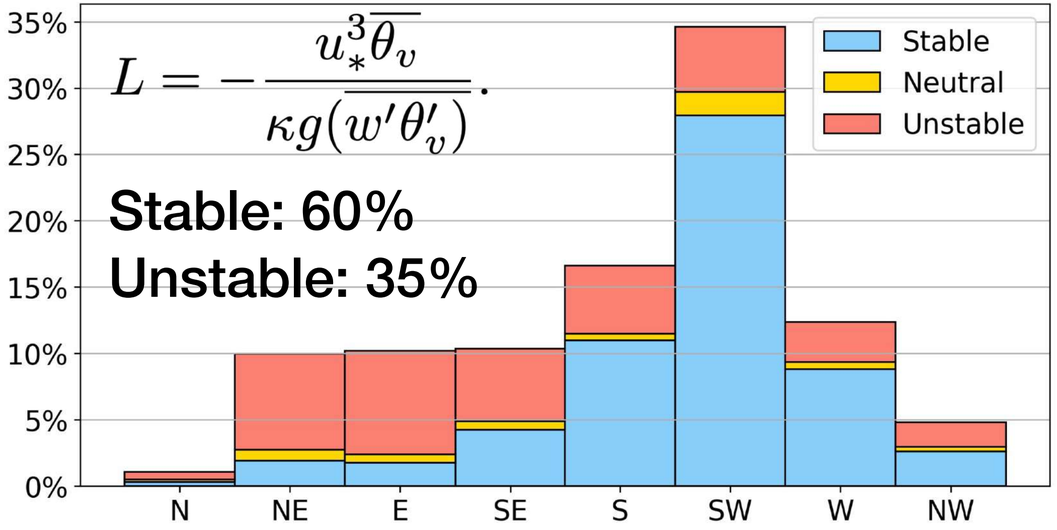
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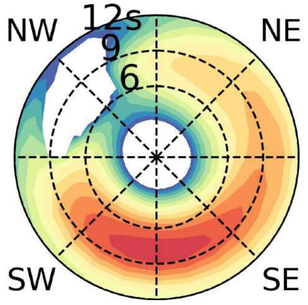


(a) Atmospheric stability conditions over MA/RI

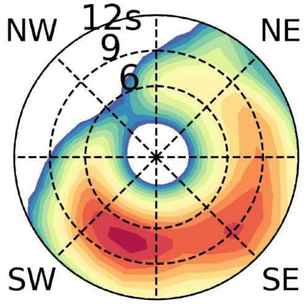


Wave

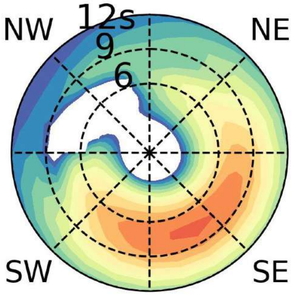
NDBC 44008



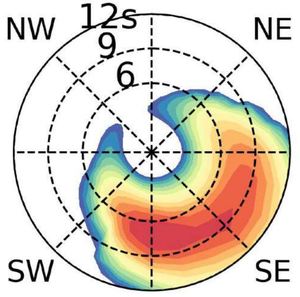
NOWF



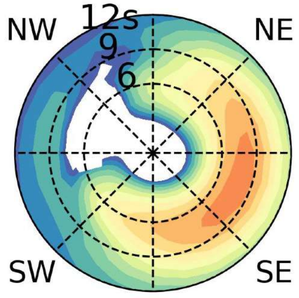
NDBC 44025



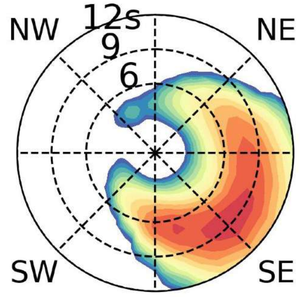
NOWF



NDBC 44009



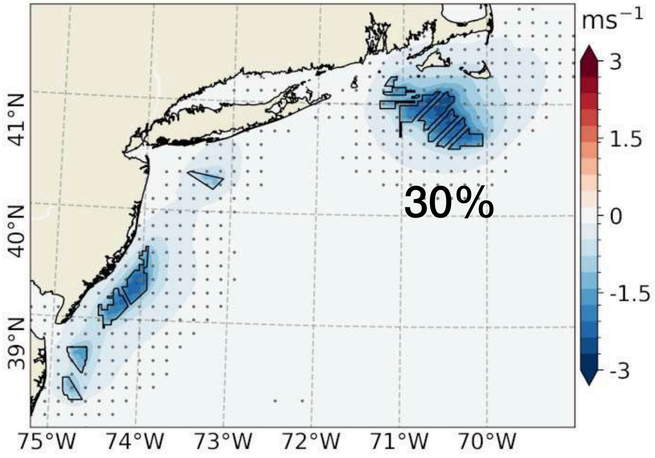
NOWF



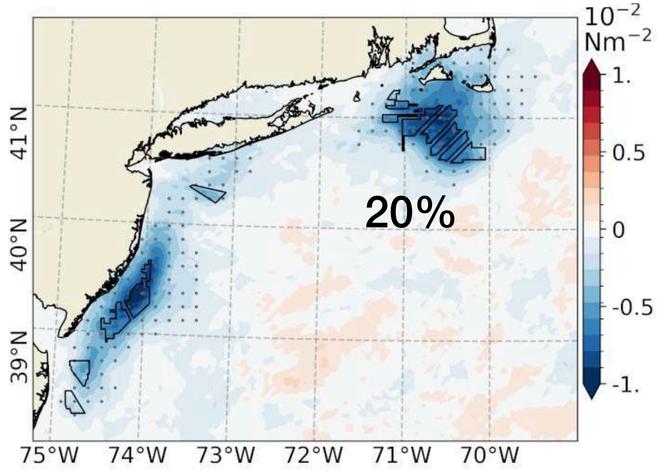


Climatological effects of wind wakes

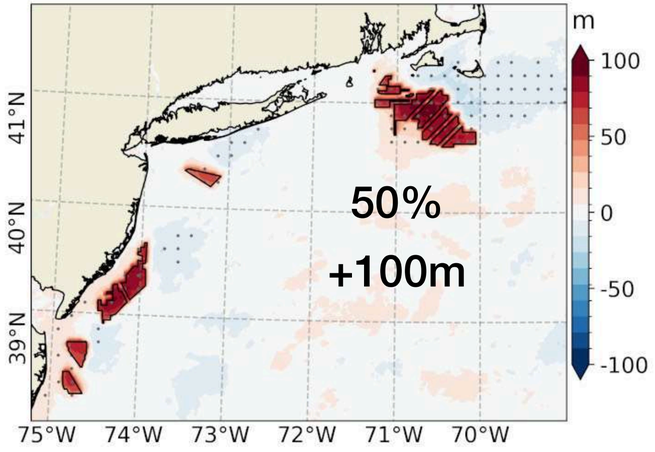
Hub height wind speeds



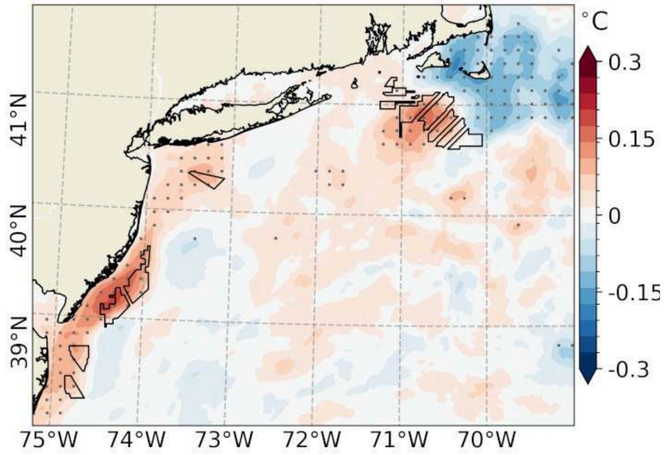
Wind stress



PBLH



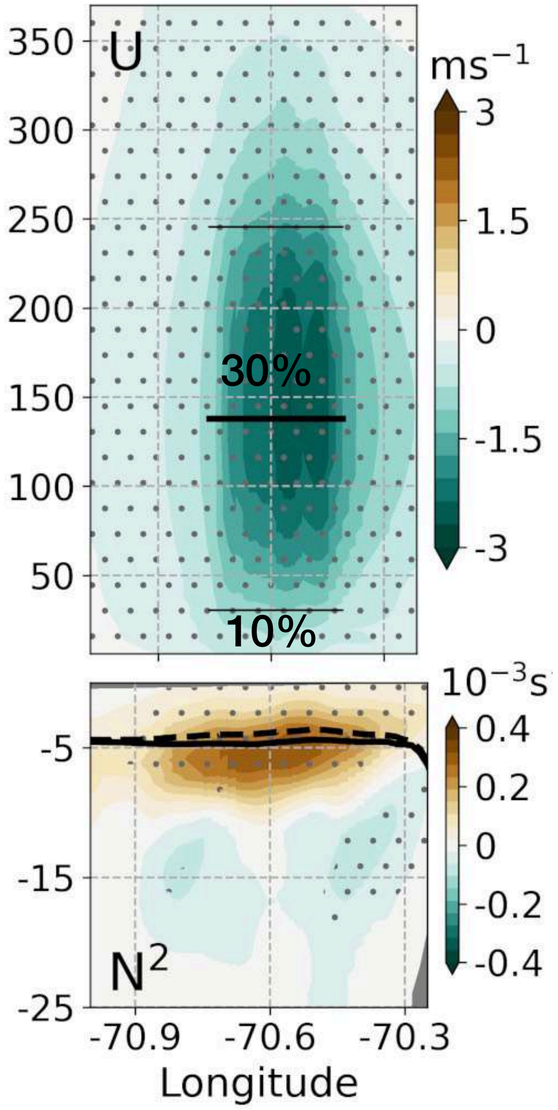
T2



dots: 99% confidence level

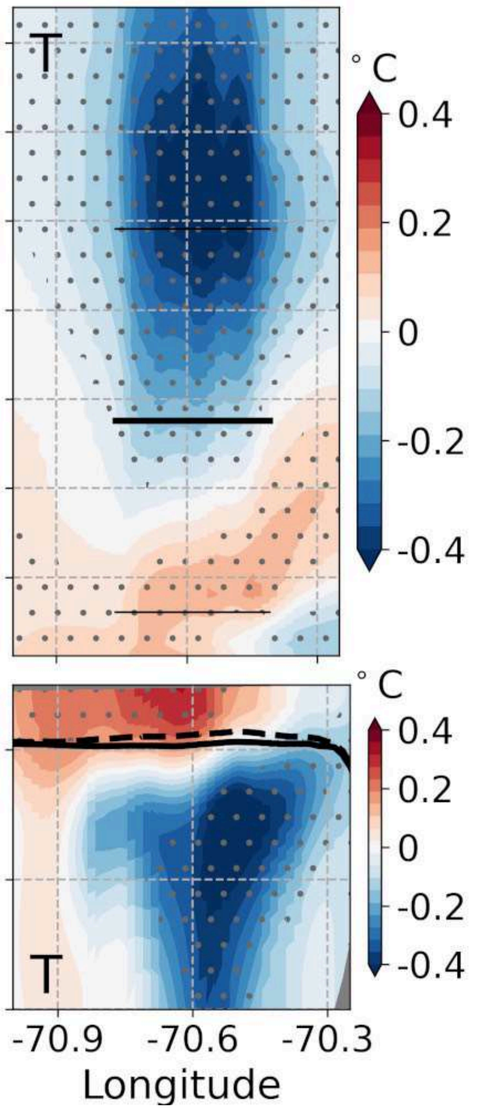
Wind speed

OC_WF-OC_NWF

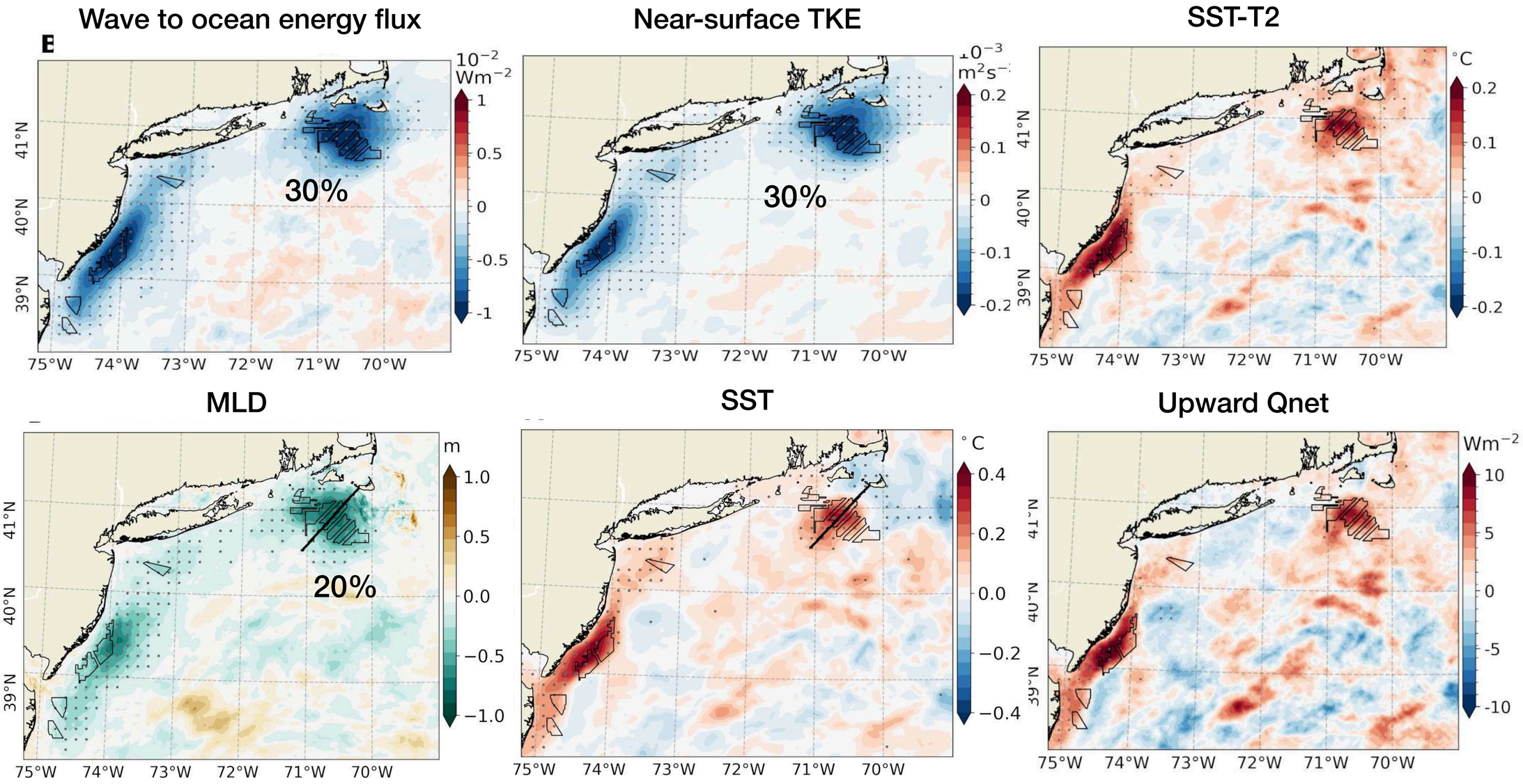


Temperature

OC_WF-OC_NWF



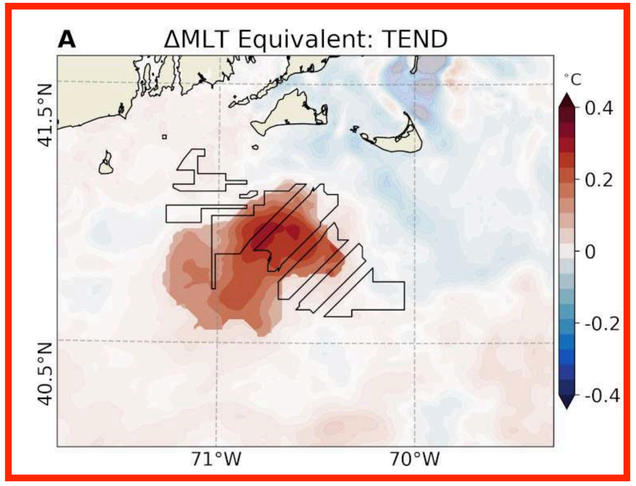
Climatological effects of wind wakes on the ocean

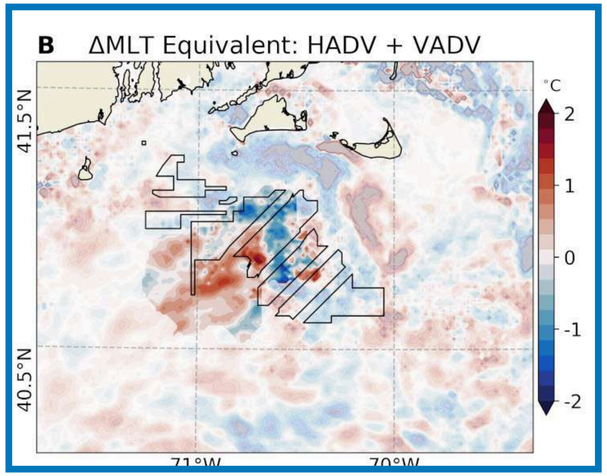


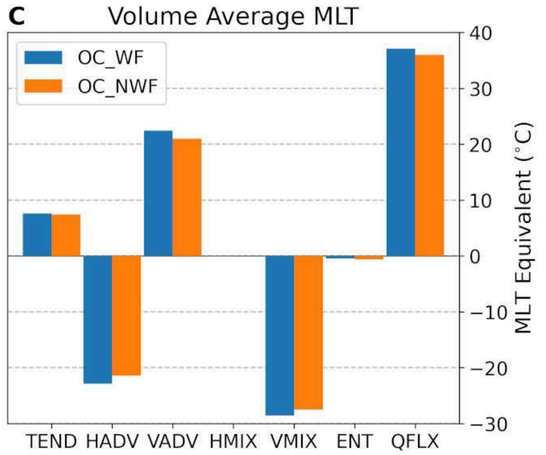
dots: 99% confidence level

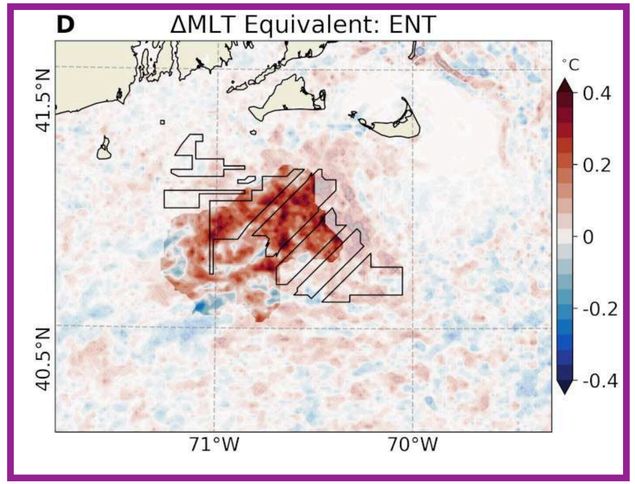
Quantifying SST warming driven by ocean ML processes

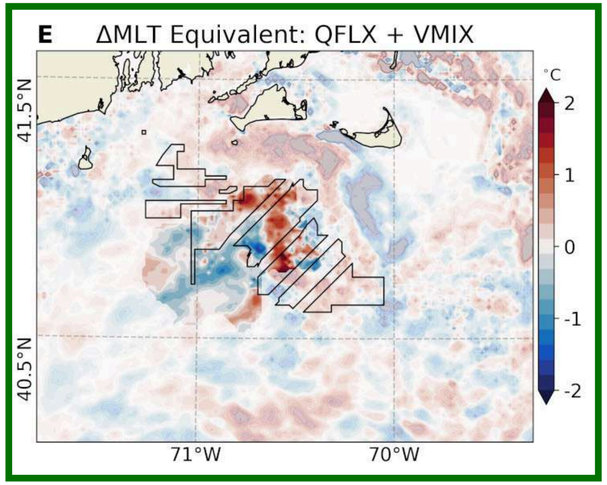
$$\underbrace{\frac{\partial \langle T \rangle}{\partial t}}_{\text{TEND}} = \underbrace{-\langle \nabla \cdot (\boldsymbol{u}T) \rangle}_{\text{HADV}} \underbrace{+\frac{1}{H}(wT)_{z=-h}}_{\text{VADV}} \underbrace{+\langle \nabla \cdot (\kappa_h \nabla T) \rangle}_{\text{HMIX}} \underbrace{-\frac{1}{H}\left(\kappa_v \frac{\partial T}{\partial z}\right)_{z=-h}}_{\text{VMIX}} \underbrace{+\frac{Q_{net} - Q_{sw}|_{z=-h}}{\rho c_p H}}_{\text{QFLX}} \underbrace{-\frac{\Delta T}{H}\frac{\partial h}{\partial t}}_{\text{ENT}}$$

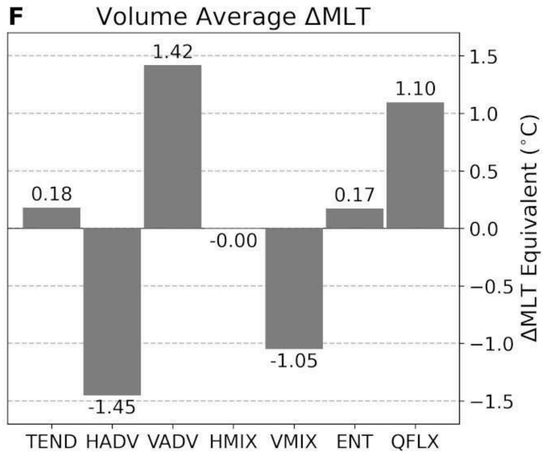












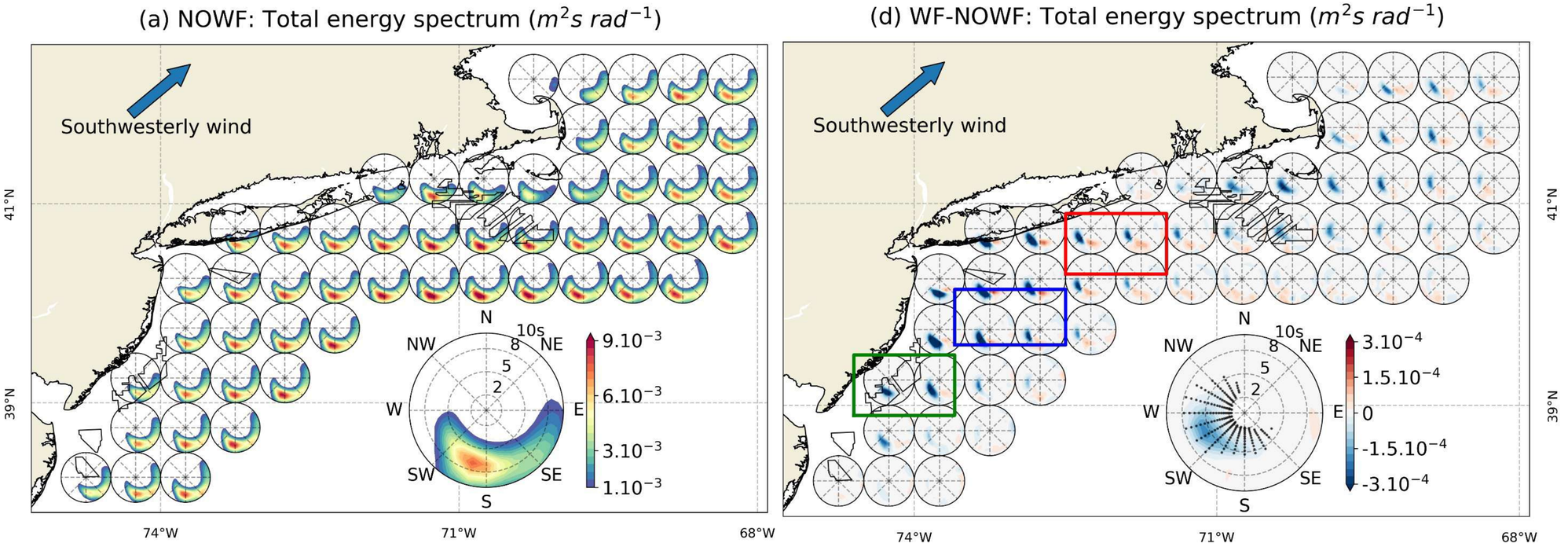
Time-integrated ROMS online heat budget:

Reduced entrainment cooling due to shallow MLD plays a dominant role.

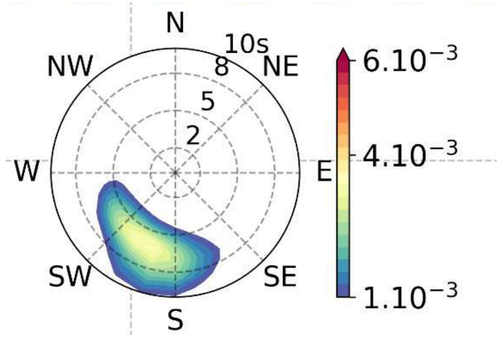
(not shown) In NJ, upwelling change and vertical mixing contribute as well.

Sauvage et al. (2025)

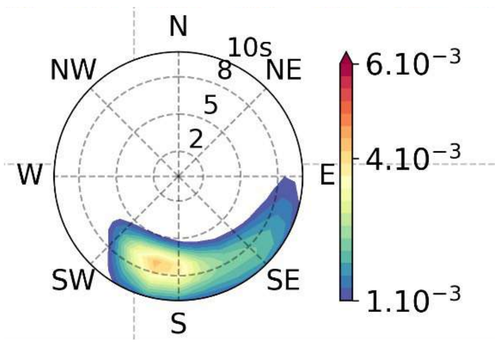
Directional Wave Spectra



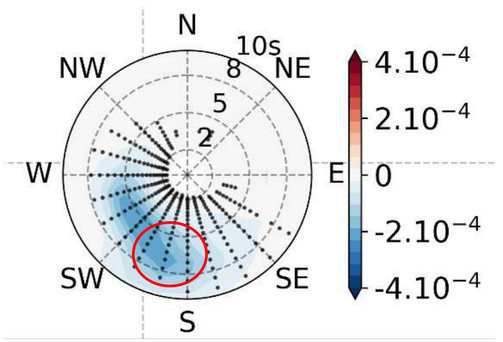
NOWF: wind-sea



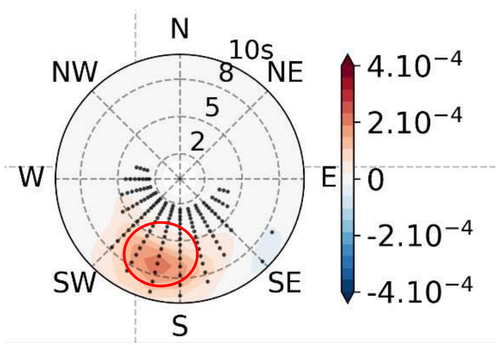
NOWF: swell



WF-NOWF: wind-sea



WF-NOWF: swell

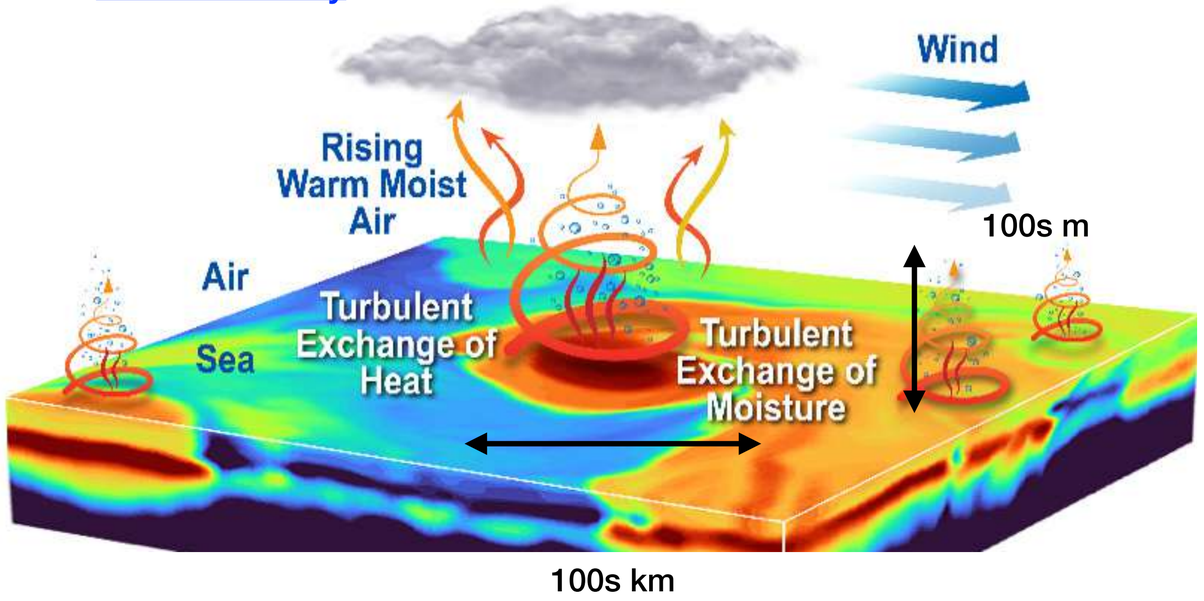


Wave regime shift from ‘wind-sea’ to ‘swell’ as the wind speed is reduced

Ocean Mesoscale Air-Sea Interaction

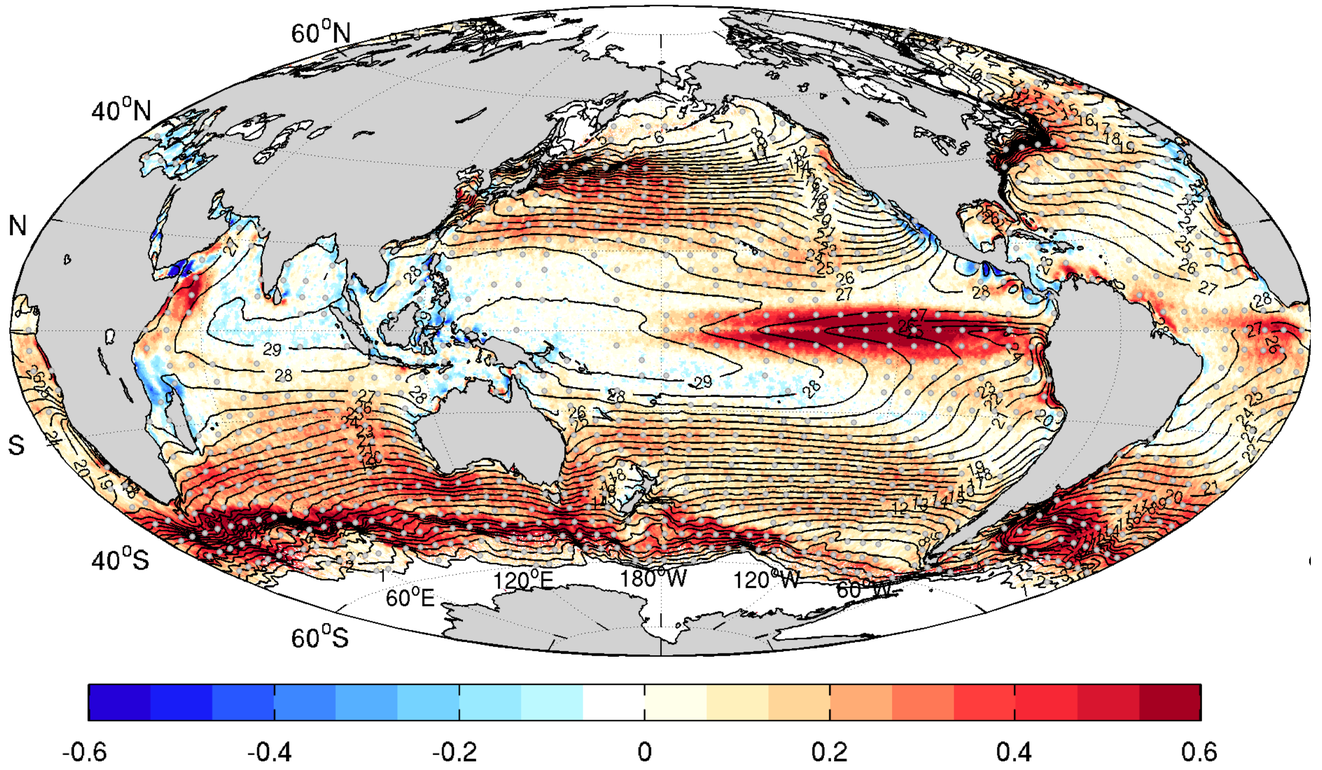
SST, current, and surface drag mediated by the ocean mesoscale eddies modifies the MABL stability, surface wind, and air-sea flux (e.g., Chelton et al. 2004; Small et al. 2008; Seo et al. 2023)

[NASA Butterfly]



Gentemann et al. (2020)

Correlation bet'n wind and SST (spatially high passed)



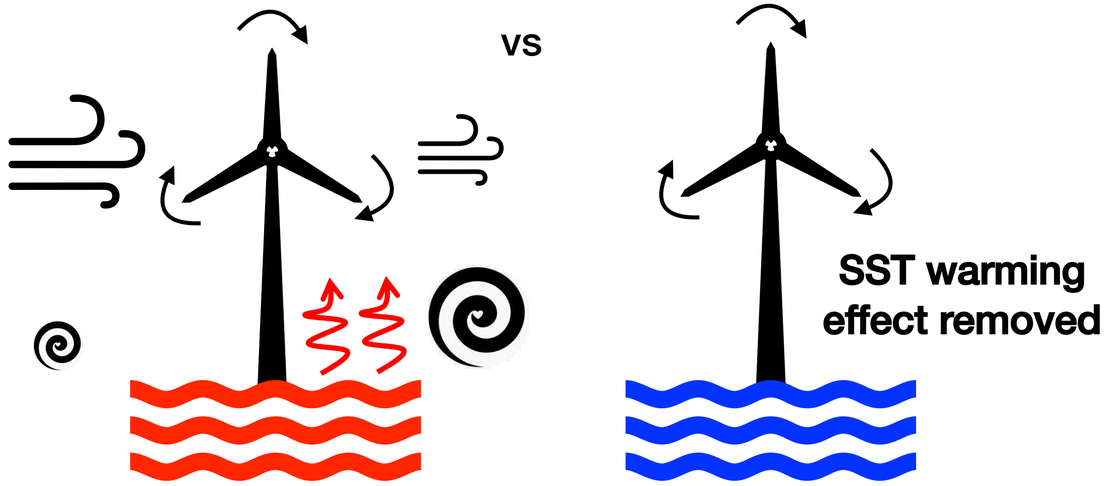
Seo (2017)

Isolating the coupled effect of offshore wind farms

OC_WF
(Wind farm and coupling)

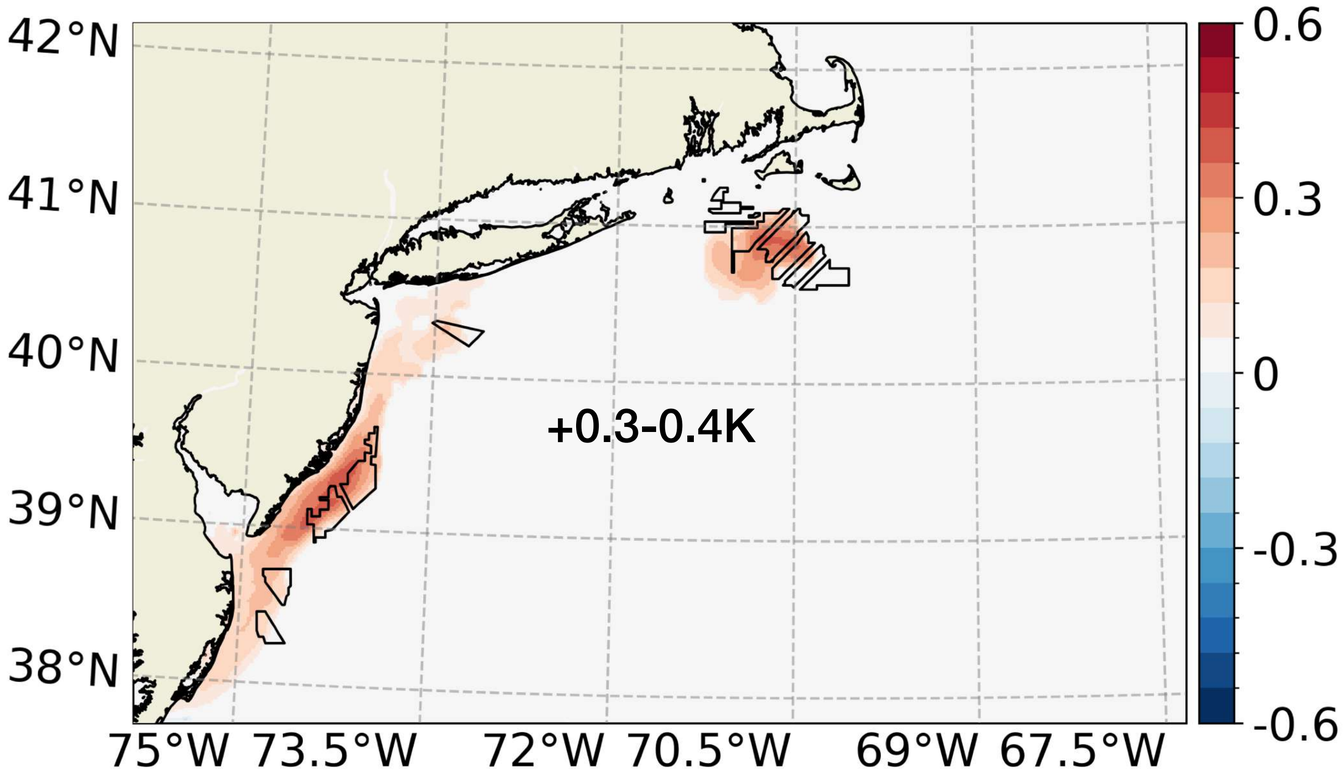
vs

NOC_WF
(Wind farm but no coupling)



Experiment	Wind Farm (WF)	Ocean Coupling (OC)
OC_WF	Yes	Yes
OC_NWF	No	Yes
NOC_WF	Yes	No

Positive SST anomaly removed in NOC_WF

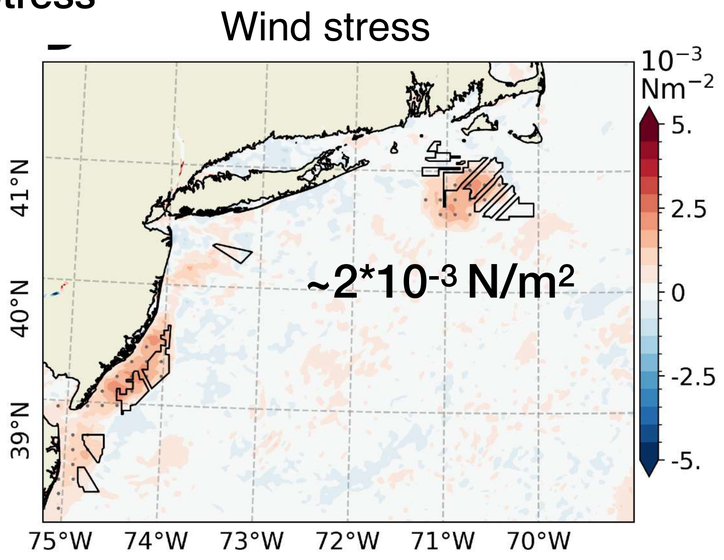


Isolating the coupled effect of offshore wind farms

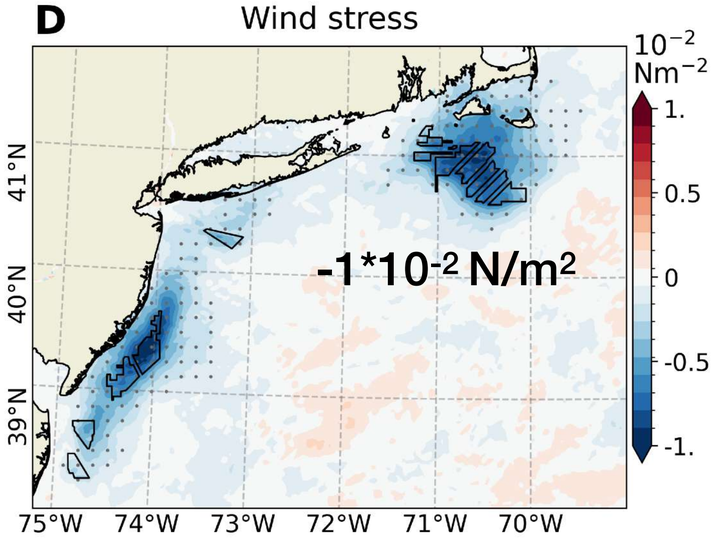
Efect of wind farm with ocean coupling

Efect of wind farm without ocean coupling

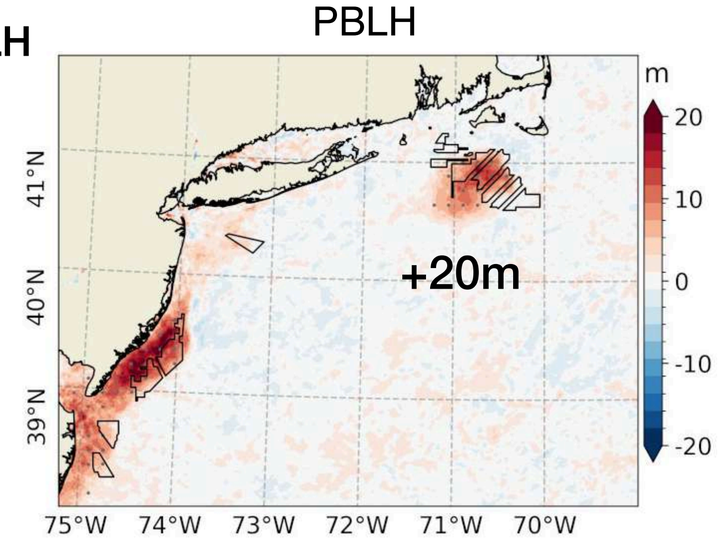
Wind Stress



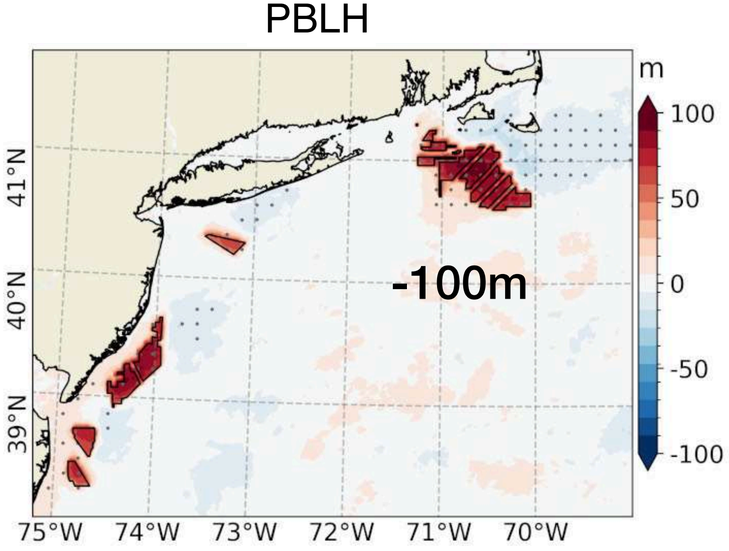
recall



PBLH



recall

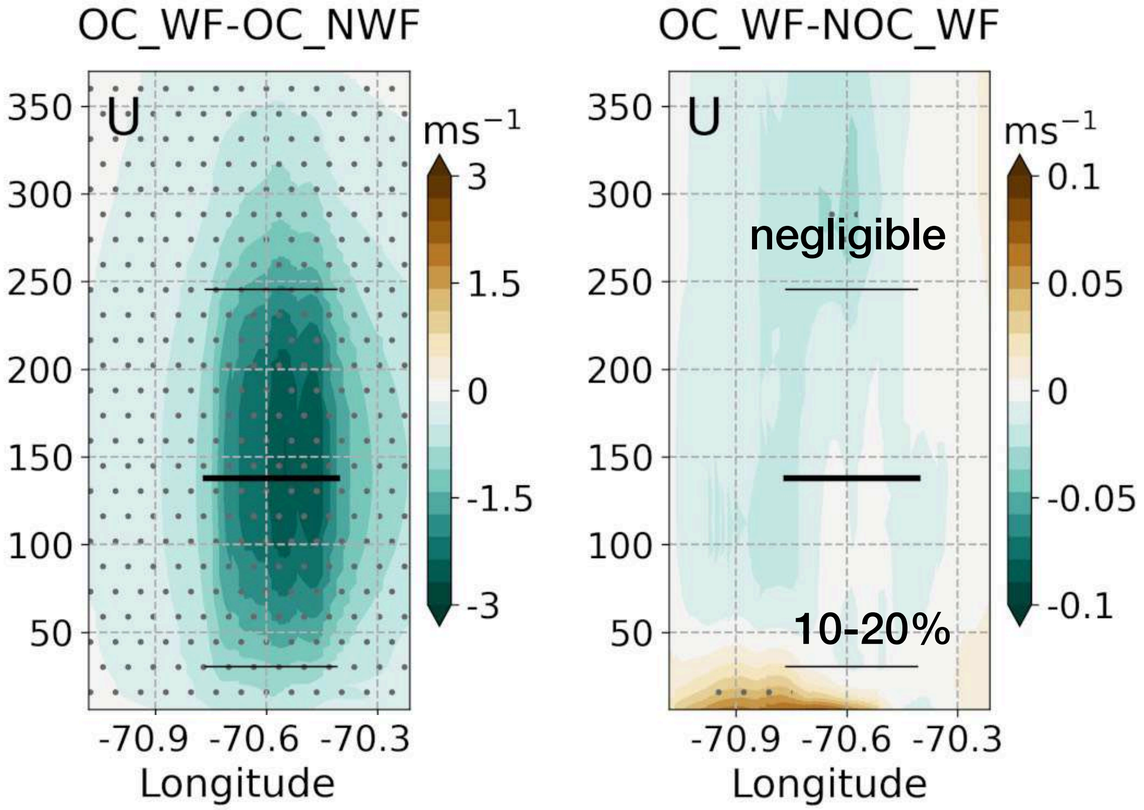


Local ocean coupling increases, offsetting wind-stress deficits and contributing to PBL deepening by 15-20%

Uncoupled models may overestimate the stress deficits and underestimate the PBLH deepening.

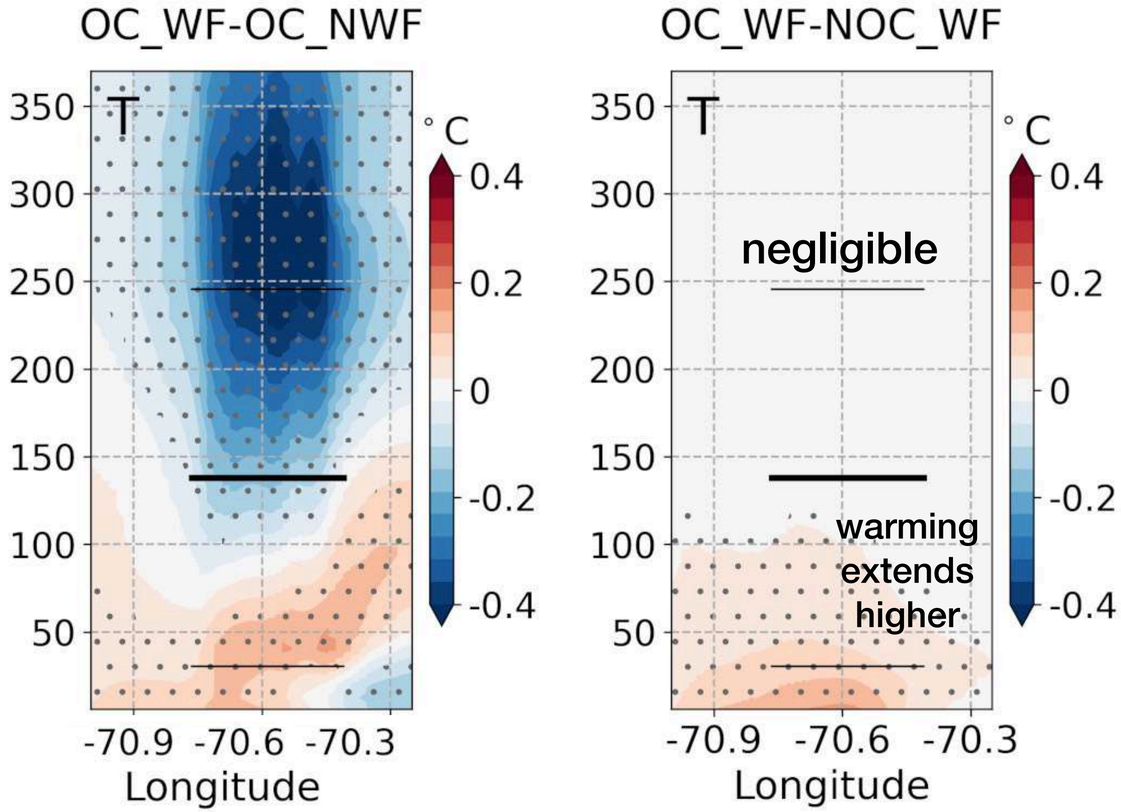
How high does the ocean coupling effect reach in the ABL?

Wind speed



The wake-driven coupling modulates the ABL stability, but its impacts on wind appear to be limited to SL (<50 m).

Temperature



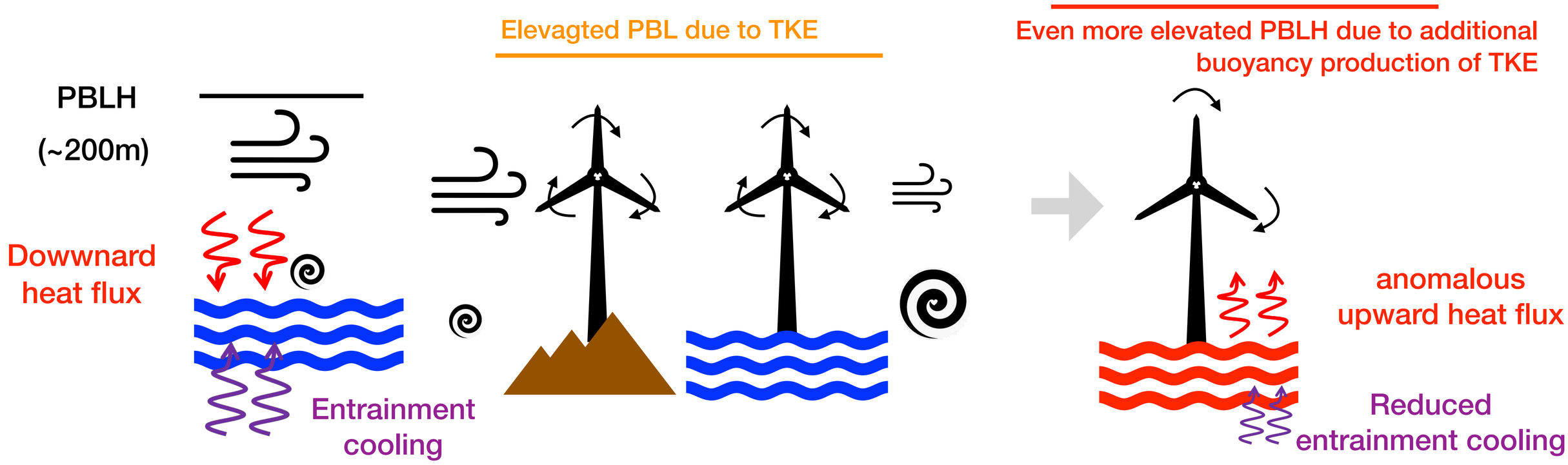
No evidence of the notable coupling effects on winds at turbine-relevant heights and hence accumulated power output.

Summary and Discussion

Unperturbed condition

Wind farm effect **without ocean coupling**

Wind farm effect **with ocean coupling**



- Wake-driven wind stress reduction leads to anomalously warm SST of 0.3-0.4°C.
- SST warming is driven primarily by the ocean mixed-layer processes (bottom-up).
- The ocean coupling effect is most pronounced in the surface layer and has no impact at turbine-relevant heights.